

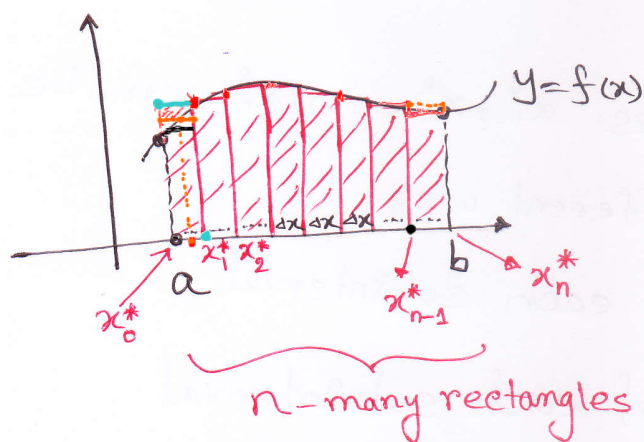
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A function f is said to be integrable on a finite closed interval $[a, b]$ if the limit exists and hence denoted as

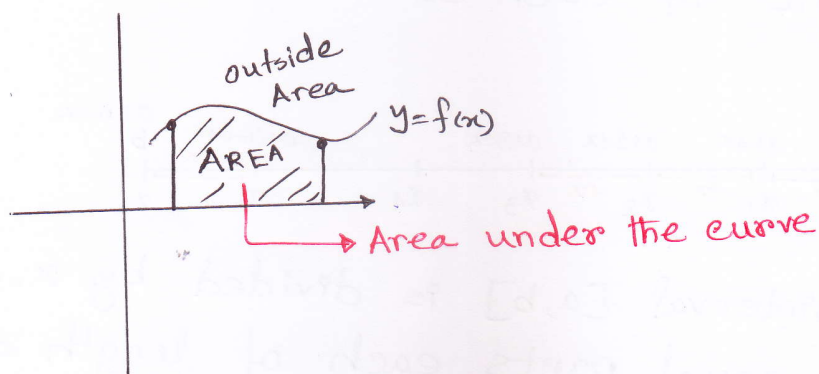
$$\text{Area} = \lim_{n \rightarrow \infty} \underbrace{\sum_{k=1}^n f(x_k^*) \Delta x}_{\text{Riemann Sum}} = \underbrace{\int_a^b \underbrace{f(x)}_{\text{integrand}} dx}_{\text{Riemann Integral}}$$

upper limit
lower limit

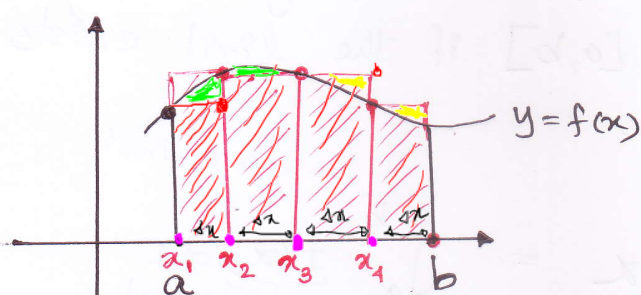


$$\Delta x = \frac{b-a}{n}$$

subintervals



Left end point approximation

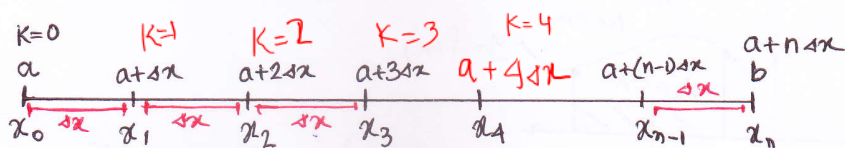


$$x_k^* = x_{k-1} = a + (k-1)\Delta x$$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n \underbrace{f(x_k^*)}_{\text{height}} \underbrace{\Delta x}_{\text{width}} \quad (*) \quad \text{From previous page}$$

In the eqn (*), the values $x_1^*, x_2^*, \dots, x_n^*$ can be chosen into three different ways

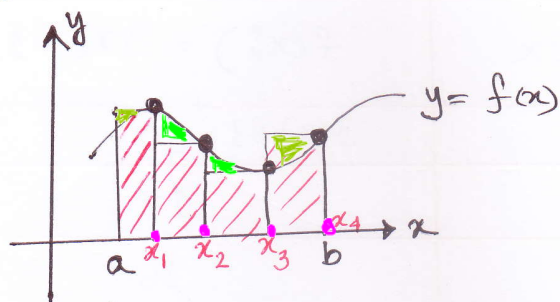
- Left end point of each subinterval
- Right end point of each subinterval
- Midpoint of each subinterval



The subinterval $[a, b]$ is divided by $x_1, x_2, \dots, x_{n-1}$ into n equal parts each of length $\Delta x = \frac{b-a}{n}$ and $x_0 = a, x_n = b$

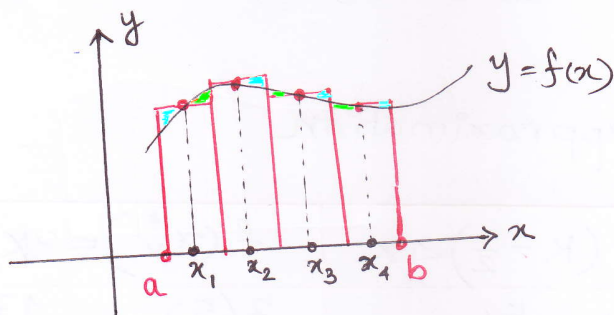
$$x_k^* = a + k\Delta x ; \quad k = 0, 1, 2, 3, 4, \dots, n$$

Right end point approximation



$$x_k^* = x_k = a + k\Delta x$$

Midpoint approximation



$$\begin{aligned} \text{Mid} &= \frac{\text{Right} + \text{Left}}{2} \\ &= \frac{k + (k-1)}{2} \\ &= k - \frac{1}{2} \end{aligned}$$

$$x_k^* = a + (k - \frac{1}{2})\Delta x$$

$$x_k^* = \frac{1}{2} (x_{k-1} + x_k)$$

average of left & right end pt approximation

$$= \frac{1}{2} \left[\underbrace{a + (k-1)\Delta x}_{\text{Left}} + \underbrace{a + k\Delta x}_{\text{Right}} \right]$$

$$= \frac{1}{2} (2a + 2k\Delta x - \Delta x)$$

$$= a + \frac{1}{2} (2k-1)\Delta x$$

$$= a + (k - \frac{1}{2})\Delta x$$

Example 1 $f(x) = 3x + 1$, $a = 2$, $b = 6$, $n = 4$

Left end point approximation

k	$x_k^* = a + (k-1)\Delta x$	$f(x_k^*) = 3x_k^* + 1$
1	$2 + (1-1)1 = 2$	$3(2) + 1 = 7$
2	$2 + (2-1)1 = 3$	$3(3) + 1 = 10$
3	$2 + (3-1)1 = 4$	13
4	$2 + (4-1)1 = 5$	16

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x = 46 \times 1 = 46$$

[3]

$$\sum_{k=1}^4 f(x_k^*) = 46$$

$$\begin{aligned} \Delta x &= \frac{b-a}{n} \\ &= \frac{6-2}{4} \\ &= \frac{4}{4} \\ &= 1 \end{aligned}$$

Right end point approximation

k	$x_k^* = a + k\Delta x$	$f(x_k^*) = 3x_k^* + 1$
1	$2 + 1(1) = 3$	$3(3) + 1 = 10$
2	$2 + 2(1) = 4$	$3(4) + 1 = 13$
3	$2 + 3(1) = 5$	$3(5) + 1 = 16$
4	$2 + 4(1) = 6$	$3(6) + 1 = 19$

$\Delta x = 1$
found in
previous
exercise

$$\sum_{k=1}^4 f(x_k^*) = 58$$

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x = 58 \times 1 = 58$$

Mid-point approximation

k	$x_k^* = a + (k - \frac{1}{2})\Delta x$	$f(x_k^*) = 3x_k^* + 1$
1	$2 + (1 - \frac{1}{2})1 = \frac{5}{2}$	$3(\frac{5}{2}) + 1 = \frac{17}{2}$
2	$2 + (2 - \frac{1}{2})1 = \frac{7}{2}$	$3(\frac{7}{2}) + 1 = \frac{23}{2}$
3	$2 + (3 - \frac{1}{2})1 = \frac{9}{2}$	$3(\frac{9}{2}) + 1 = \frac{29}{2}$
4	$2 + (4 - \frac{1}{2})1 = \frac{11}{2}$	$3(\frac{11}{2}) + 1 = \frac{35}{2}$

$\Delta x = 1$

$$\sum_{k=1}^4 f(x_k^*) = \frac{104}{2} = 52$$

$$A = \sum_{k=1}^4 f(x_k^*) \Delta x = 52 \times 1 = 52$$

Alternatively: Mid pt approximation = Average of left & right end pt approximation

$$= \frac{1}{2} (46 + 58) = 52$$

Theorem

If the function f is continuous on $[a, b]$, [n is unknown or extremely large] then the net signed area A between $y = f(x)$ and interval $[a, b]$ is defined by $A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$

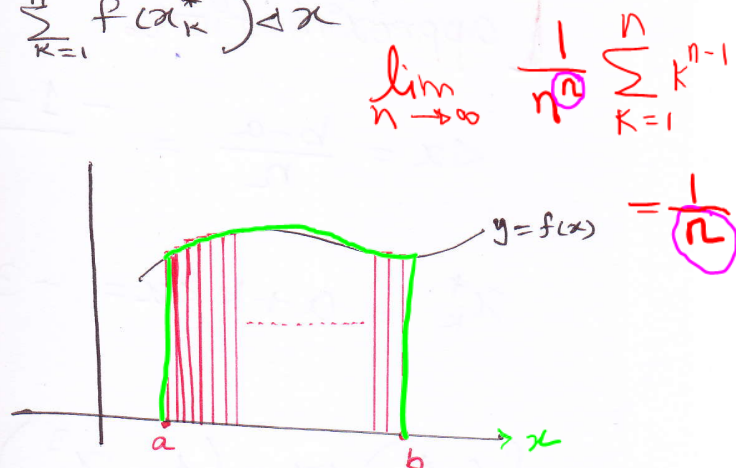
where

$$a) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1 = 1 = \frac{1}{1}$$

$$b) \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k = \frac{1}{2}$$

$$c) \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k^2 = \frac{1}{3}$$

$$d) \lim_{n \rightarrow \infty} \frac{1}{n^4} \sum_{k=1}^n k^3 = \frac{1}{4}$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^6} \sum_{k=1}^n k^5 = ? = \frac{1}{6}$$

→ We can see the pattern of the theorem above. Anything that does not match with the above pattern will produce '0' as a result.

Ex $\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k = 0$ (False)

Ex Find the area under the curve $y = 1 - x^3$ over the interval $[-3, -1]$

[We can use left end, right end or midpoint approximation.]

width $\rightarrow \Delta x = \frac{b-a}{n} = \frac{-1 - (-3)}{n} = \frac{2}{n}$ $\because [-3, -1] = [a, b]$

input to find height $\rightarrow x_k^* = a + k \Delta x = -3 + k \left(\frac{2}{n}\right) = -3 + \frac{2k}{n}$ using right end point approximation

height $\rightarrow f(x_k^*) \Delta x = (1 - x_k^3) \frac{2}{n}$
 $= \left[1 - \left(-3 + \frac{2k}{n}\right)^3 \right] \frac{2}{n}$

$$= \left[1 - \left(-27 + 3(-3)^2 \left(\frac{2k}{n}\right) + 3(-3) \left(\frac{2k}{n}\right)^2 + \left(\frac{2k}{n}\right)^3 \right) \right] \frac{2}{n}$$

$$= \left[1 - \left(-27 + \frac{54k}{n} - \frac{36k^2}{n} + \frac{8k^3}{n^3}\right) \right] \frac{2}{n}$$

$$= \left(1 + 27 - \frac{54k}{n} + \frac{36k^2}{n} - \frac{8k^3}{n^3} \right) \frac{2}{n}$$

$$f(x_k^*) \Delta x = \frac{56}{n} - \frac{108k}{n^2} + \frac{72k^2}{n^3} - \frac{16k^3}{n^4}$$

$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{1}{n} (56) - \frac{1}{n^2} k (108) + \frac{1}{n^3} k^2 (72) - \frac{1}{n^4} k^3 (16) \right]$

$\uparrow$
Area

$$= (1)56 - \left(\frac{1}{2}\right)108 + \left(\frac{1}{3}\right)72 - \left(\frac{1}{4}\right)16 = 22$$

[6]

Chapter 5.2, 5.3 Integration & Antiderivative Week 1
Anton's Calculus

A function 'F' is called an antiderivative of the function 'f' on a given interval I if

$$\boxed{F'(x) = f(x)} \text{ for all } x \text{ in } I \text{ or } \boxed{\int f(x) dx = F(x)}$$

Ⓛ⓪⓲Ⓛ⓪⓲

Ⓛ⓪⓲ & Ⓛ⓪⓲ represents same fact with different notation

$$F'(x) = f(x) \text{ --- } \textcircled{i}$$

$$\Rightarrow \frac{dF(x)}{dx} = f(x)$$

$$\Rightarrow dF(x) = f(x) dx \Rightarrow \int dF(x) = \int f(x) dx$$

$$F(x) = \int f(x) dx \text{ --- } \textcircled{ii}$$

if we differentiate an antiderivative of $f(x)$, we obtain $f(x)$ back again.

$$\frac{d}{dx} \left[\int \textcircled{f(x)} dx \right] = f(x) + C$$

↓
integrand

↓
constant of integration

Some Fundamental Theorems of integral calculus:

i) If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$ then

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

integration formula
available at chapter 7.1 of
Calculus 10 Ed. by Anton

ex $\int_{\sqrt{2}}^2 \frac{dx}{x\sqrt{x^2-1}}$

$$= \int_{\sqrt{2}}^2 \frac{dx}{x\sqrt{x^2-1}}$$

$$= \left[\frac{1}{1} \sec^{-1} \frac{x}{1} \right]_{\sqrt{2}}^2$$

$$= \sec^{-1}(2) - \sec^{-1}(\sqrt{2})$$

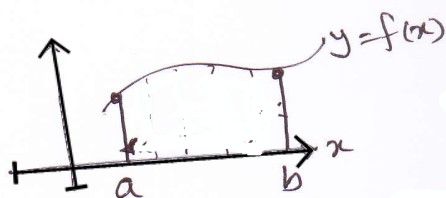
formula:

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{u}{a} \right|$$

Mean Value Theorem

ii) If f is continuous on a closed interval $[a, b]$, then there is at least one number x^* in $[a, b]$ such that

$$\int_a^b f(x) dx = f(x^*) (b-a)$$



Indefinite & Definite Integral Week 1

Some properties of integration

$$\textcircled{i} \int_a^a f(x) dx = 0 \quad \textcircled{ii} \int_a^b f(x) dx = - \int_b^a f(x) dx$$

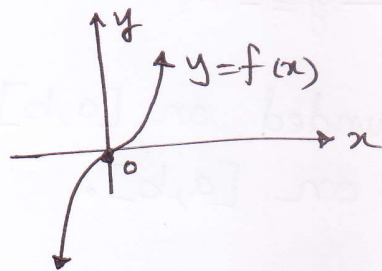
$$\textcircled{iii} \int_a^b c f(x) dx = c \int_a^b f(x) dx$$

$$\textcircled{iv} \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) \pm \int_a^b g(x) dx$$

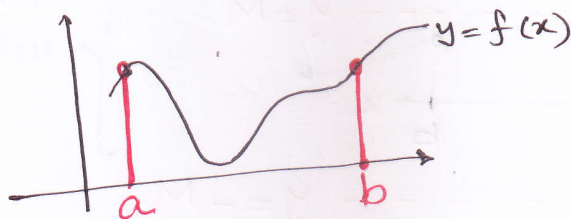
$$\textcircled{v} \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

while $f(x)$ is piecewise smooth curve

Ex $y = x^3$ is concave down on $(-\infty, 0)$ & concave up on $(0, +\infty)$. '0' is the inflection pt.

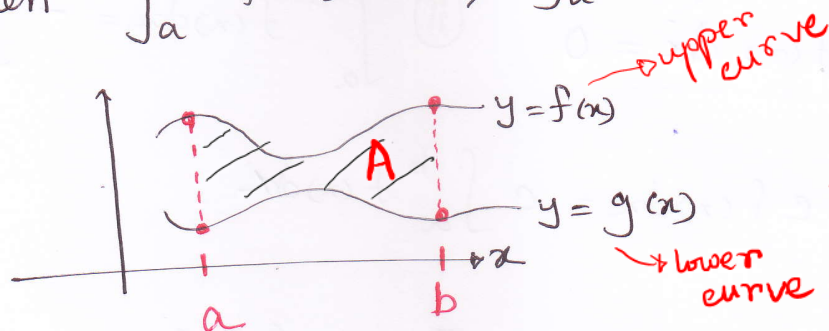


(vi) If $f(x) \geq 0$, for all $x \in [a, b]$ then $\int_a^b f(x) dx \geq 0$



(vii) If $f(x) \geq g(x)$ for all $x \in [a, b]$

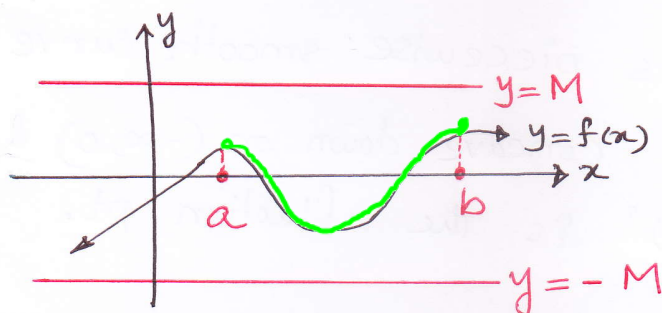
then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$



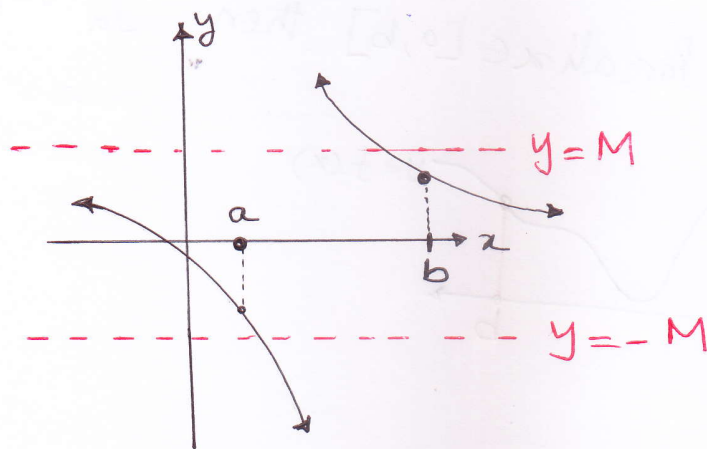
$A = \text{area between curves}$

$$= \int_a^b [f(x) - g(x)] dx$$

(viii) If $f(x)$ is bounded, for example $-M \leq f(x) \leq M$ for all $x \in [a, b]$, then $f(x)$ is integrable.



(ix) If $f(x)$ is not bounded on $[a, b]$ then $f(x)$ is not integrable on $[a, b]$.



Here $f(x)$ is unbounded on $[a, b]$.

Examples

Integration by Substitution

$$\textcircled{1} \int \sin 2x \cos x dx$$

$$= \int \frac{1}{2} [\sin(2x+x) + \sin(2x-x)] dx$$

$$= \frac{1}{2} \int (\sin 3x + \sin x) dx$$

$$= \frac{1}{2} \left[-\frac{\cos 3x}{3} - \cos x \right] + C$$

$$\sin a \cos b = \frac{1}{2} [\sin(a+b) + \sin(a-b)]$$

$$\textcircled{2} \int \frac{dx}{\sqrt{1-4x^2}}$$

$$= \int \frac{dx}{\sqrt{1-(2x)^2}}$$

let

$$2x = z$$

$$2dx = dz$$

$$dx = \frac{1}{2} dz$$

$$= \frac{1}{2} \int \frac{dz}{\sqrt{1-z^2}}$$

$$= \frac{1}{2} \sin^{-1}(z) + C$$

$$= \frac{1}{2} \sin^{-1}(2x) + C$$

We know

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$$

$$\textcircled{3} \int x \sqrt{1-x} dx$$

$$= \int (1-z) \sqrt{z} (-dz)$$

$$= - \int (\sqrt{z} - z\sqrt{z}) dz$$

$$= \int (z\sqrt{z} - \sqrt{z}) dz$$

$$= \int (z^{3/2} - z^{1/2}) dz = \frac{z^{3/2+1}}{\frac{3}{2}+1} - \frac{z^{1/2+1}}{\frac{1}{2}+1} + C = \frac{2}{5} z^{5/2} - \frac{2}{3} z^{3/2} + C$$

$$= \frac{2}{5} (1-x)^{5/2} - \frac{2}{3} (1-x)^{3/2} + C$$

[5]

$$\boxed{4} \int x^2 e^{-2x^3} dx$$

$$= \frac{1}{6} \int e^{-z} dz$$

$$= \frac{1}{6} \left[\frac{e^{-z}}{-1} \right] + C$$

$$= -\frac{1}{6} e^{-z} + C = -\frac{1}{6} e^{-2x^3} + C$$

$$\text{let } 2x^3 = z$$

$$6x^2 dx = dz$$

$$x^2 dx = \frac{1}{6} dz$$

$$\boxed{5} \int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx$$

let

$$e^x - e^{-x} = z$$

$$(e^x + e^{-x}) dx = dz$$

$$= \int \frac{dz}{z}$$

$$= \ln z + C = \ln(e^x - e^{-x}) + C$$

$$\boxed{6} \int \frac{e^x}{1 + e^{2x}} dx$$

let

$$e^x = z$$

$$e^x dx = dz$$

$$= \int \frac{e^x}{1 + (e^x)^2} dx$$

$$\int \frac{1}{1+z^2} dz = \tan^{-1} z$$

$$= \int \frac{dz}{1+z^2} = \tan^{-1} z + C = \tan^{-1}(e^x) + C$$

$$\boxed{7} \int \frac{\sec^2 x \, dx}{\sqrt{1 - \tan^2 x}}$$

$$\begin{aligned} \tan x &= z \\ \sec^2 x \, dx &= dz \end{aligned}$$

$$= \int \frac{dz}{\sqrt{1 - z^2}} = \sin^{-1} z + C = \sin^{-1}(\tan x) + C$$

$$\boxed{8} \int \frac{\sin x}{\cos^2 x + 1} \, dx$$

Hint

$$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x$$

substitute $\cos x = z$

$$= -\tan^{-1}(\cos x) + C$$

$$\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}} \, dx$$

$$= 2 \int \sec^2 z \, dz$$

$$= 2 \tan(z) + C$$

$$= 2 \tan \sqrt{x} + C$$

let

$$\sqrt{x} = z$$

$$\frac{1}{2\sqrt{x}} \, dx = dz$$

$$\frac{1}{\sqrt{x}} \, dx = 2 \, dz$$

7.8 Infinite Integrals (Improper Integrals)

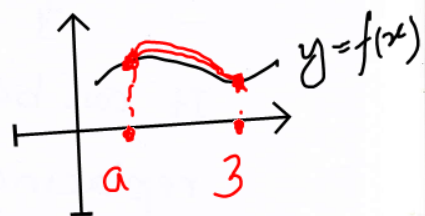
Week 1

Anton's
Calculus
10th Ed.

Consider the followings:

$$\int_c^\infty (ax+b) dx ; \int_{-\infty}^c (ax+b) dx$$

$$\int_{-\infty}^\infty (ax+b) dx ; \int_a^3 \frac{1}{x-3} dx$$



An improper integral is a definite integral that has either or both limits infinite or an integrand that approaches infinitely at one or more points in the range of integration.

$$\int_{\text{Range of integration}} f(x) dx = F(x) \begin{cases} \rightarrow \text{Converge} \\ \quad \text{(If the solution exists)} \\ \rightarrow \text{Diverge} \\ \quad \text{(If the solution does not exist)} \end{cases}$$

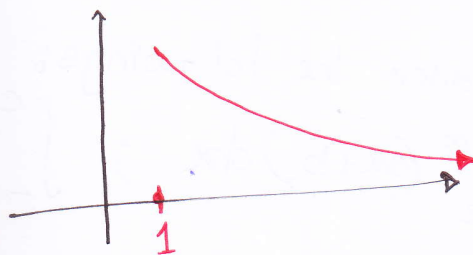
Labels in the diagram:
- An arrow points from the text "Range of integration" to the integral symbol $\int$.
- An arrow points from the text "Integrand" to the function $f(x)$ inside the integral.

It cannot be computed using normal integral.
So we introduce limit.

Consider:

$$\int_1^{\infty} x^{-2} dx$$

$$y = \frac{1}{x^2}, \quad x \in [1, \infty)$$



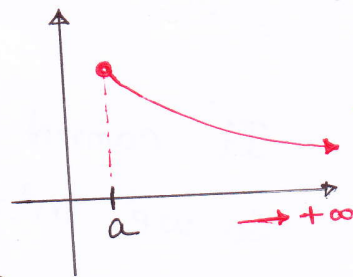
It can be computed by replacing infinite limits with finite values:

$$\begin{aligned} \int_1^{\infty} x^{-2} dx &= \lim_{l \rightarrow \infty} \int_1^l x^{-2} dx = \lim_{l \rightarrow \infty} \left[\frac{x^{-2+1}}{-2+1} \right]_1^l \\ &= \lim_{l \rightarrow \infty} \left[\frac{-1}{x} \right]_1^l \\ &= \lim_{l \rightarrow \infty} \left[-\frac{1}{l} + 1 \right] \\ &= -\frac{1}{\infty} + 1 = 0 + 1 = 1 \end{aligned}$$

Observe few cases:

1(a) The improper integral of 'f' over the interval $[a, +\infty)$ is defined as:

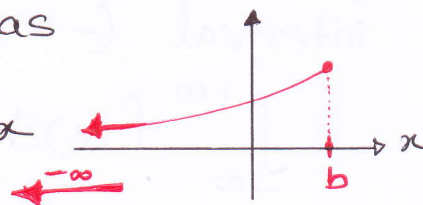
$$\int_a^{+\infty} f(x) dx = \lim_{l \rightarrow +\infty} \int_a^l f(x) dx$$



The integral is said to converge if the limit exists and diverge if it does not.

1(b) The improper integral of f over the interval $(-\infty, b]$ is defined as

$$\int_{-\infty}^b f(x) dx = \lim_{K \rightarrow -\infty} \int_K^b f(x) dx$$



The integral is said to converge if the limit exists and diverge if it does not.

Example $\int_4^{+\infty} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$

$$= \lim_{l \rightarrow +\infty} \int_4^l \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$$

$$= \lim_{l \rightarrow +\infty} \left[\ln(x-1) - \ln(x+1) \right]_4^l$$

$$= \lim_{l \rightarrow +\infty} \left[\ln \left(\frac{x-1}{x+1} \right) \right]_4^l$$

$$= \lim_{l \rightarrow +\infty} \left[\ln \left(\frac{l-1}{l+1} \right) - \ln \left(\frac{3}{5} \right) \right]$$

$$= \lim_{l \rightarrow +\infty} \ln \frac{l-1}{l+1} - \lim_{l \rightarrow +\infty} \ln \left(\frac{3}{5} \right)$$

$$= \lim_{l \rightarrow +\infty} \ln \frac{l+1-2}{l+1} - \lim_{l \rightarrow \infty} \ln \left(\frac{3}{5} \right)$$

$$= \lim_{l \rightarrow \infty} \ln \left(1 - \frac{2}{l+1} \right) - \ln \left(\frac{3}{5} \right)$$

$$= \ln \left(1 - \frac{2}{\infty+1} \right) - \ln \left(\frac{3}{5} \right)$$

$$= \ln(1-0) - \ln \left(\frac{3}{5} \right) = 0 - \ln \left(\frac{3}{5} \right) = \ln \left(\frac{5}{3} \right)$$

$\ln 1 = 0$

Converges.

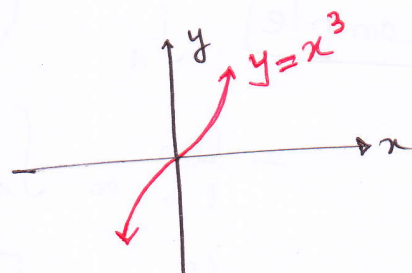
2 The improper integral of f over the interval $(-\infty, +\infty)$ is defined as

$$\int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{+\infty} f(x) dx$$

piecewise continuous function from $(-\infty, c)$ & $(c, +\infty)$.

where 'c' is any real number. The improper integral is said to converge if both terms converge and diverge if either term diverges.

Example $\int_{-\infty}^{+\infty} x^3 dx$



$$= \int_{-\infty}^0 x^3 dx + \int_0^{+\infty} x^3 dx$$

$$= \lim_{K \rightarrow -\infty} \int_K^0 x^3 dx + \lim_{l \rightarrow +\infty} \int_0^l x^3 dx$$

$$= \lim_{K \rightarrow -\infty} \left[\frac{x^4}{4} \right]_K^0 + \lim_{l \rightarrow +\infty} \left[\frac{x^4}{4} \right]_0^l$$

$$= \lim_{K \rightarrow -\infty} \left[0 - \frac{K^4}{4} \right] + \lim_{l \rightarrow +\infty} \left[\frac{l^4}{4} - 0 \right]$$

$$= -\frac{1}{4} \lim_{K \rightarrow -\infty} K^4 + \frac{1}{4} \lim_{l \rightarrow +\infty} l^4$$

$$= \frac{1}{4} \left[\lim_{l \rightarrow +\infty} l^4 - \lim_{K \rightarrow -\infty} K^4 \right]$$

$$= \frac{1}{4} [\infty - \infty] = \infty \text{ diverges.}$$

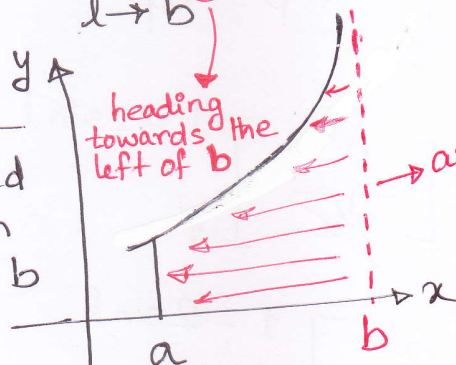
4

3 If f is continuous on the interval $[a, b]$ except for an infinite discontinuity at b , then the improper integral of f over the interval $[a, b]$ is defined as

$$\int_a^b f(x) dx = \lim_{l \rightarrow b^-} \int_a^l f(x) dx$$

The integral converges if the limit exists and diverges if it does not.

The required region is on the left of b



asymptote
(It means the curve will not touch this line or exit this line)

Example $\int_0^{\pi/6} \frac{\cos x}{\sqrt{1-2\sin x}} dx$

$$= \lim_{l \rightarrow \frac{\pi}{6}^-} \int_0^l \frac{\cos x}{\sqrt{1-2\sin x}} dx$$

$$= \lim_{l \rightarrow \frac{\pi}{6}^-} \int_0^{\sin l} \frac{dz}{\sqrt{1-2z}}$$

Observe
 $x \neq \pi/6 \quad \therefore \sin \frac{\pi}{6} = \frac{1}{2}$

$$\therefore \sqrt{1-2\sin x} = \sqrt{1-2\sin \frac{\pi}{6}}$$

$$= \sqrt{1-2\left(\frac{1}{2}\right)}$$

$$= \sqrt{1-1}$$

$$= 0.$$

$$\therefore f(x) = \frac{\cos x}{0} = \infty$$

Let $\sin x = z$
 $\cos x dx = dz$

limits:

$$x=0 \rightarrow z = \sin 0 = 0$$

$$x=l \rightarrow z = \sin l$$

$$= \lim_{l \rightarrow \frac{\pi}{6}^-} \int_0^{\sin l} \frac{dz}{\sqrt{1-2z}}$$

$$= \lim_{l \rightarrow \frac{\pi}{6}^-} \int_1^{1-2\sin l} -\frac{1}{2} \frac{dp}{\sqrt{p}}$$

$$= -\frac{1}{2} \lim_{l \rightarrow \frac{\pi}{6}^-} \int_1^{1-2\sin l} p^{-1/2} dp$$

$$= -\frac{1}{2} \lim_{l \rightarrow \frac{\pi}{6}^-} \left[\frac{p^{1/2}}{1/2} \right]_1^{1-2\sin l}$$

$$= -\lim_{l \rightarrow \frac{\pi}{6}^-} \left[p^{1/2} \right]_1^{1-2\sin l}$$

$$= -\lim_{l \rightarrow \frac{\pi}{6}^-} \left[\sqrt{1-2\sin l} - \sqrt{1} \right]$$

$$= -\sqrt{1-2\left(\frac{1}{2}\right)} + 1$$

$$= 1$$

Converges

$$\text{let } 1-2z=p$$

$$-2dz = dp$$

$$dz = -\frac{1}{2} dp$$

limits

$$z=0 \rightarrow p=1$$

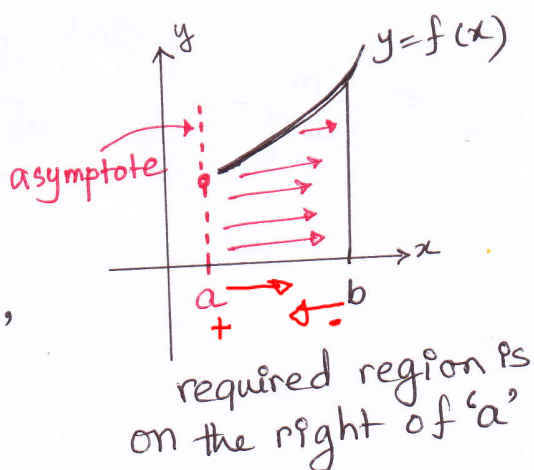
$$z=\sin l \rightarrow p=1-2\sin l$$

4 If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at a , then the improper integral of ' f ' over the interval $[a, b]$ is:

$$\int_a^b f(x) dx = \lim_{k \rightarrow a^+} \int_k^b f(x) dx$$

The integral converges if the limit exists & diverges if it does not.

[6]



Example $\int_3^4 \frac{dx}{(x-3)^2}$

$x \neq 3$
otherwise $f(x)$ is undefined.

$$= \lim_{l \rightarrow 3^+} \int_l^4 \frac{dx}{(x-3)^2}$$

$$\downarrow$$

$$\frac{1}{(x-3)^2} = \frac{1}{(3-3)^2}$$

$$= \frac{1}{0} = \infty$$

$$= \lim_{l \rightarrow 3^+} \int_{l-3}^1 \frac{dz}{z^2}$$

let $x-3=z$
 $dx=dz$

limits:
 $x=l \rightarrow z=l-3$
 $x=4 \rightarrow z=1$

$$= \lim_{l \rightarrow 3^+} \int_{l-3}^1 z^{-2} dz$$

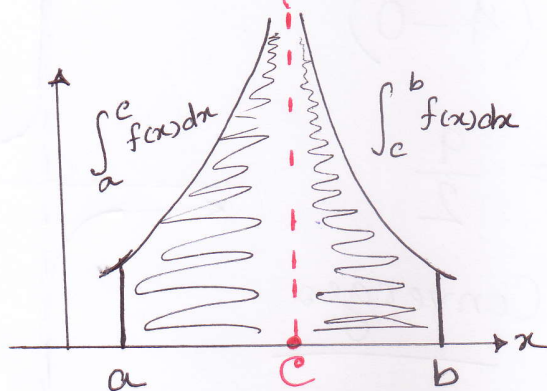
$$= \lim_{l \rightarrow 3^+} \left[\frac{z^{-1}}{-1} \right]_{l-3}^1$$

$$= - \lim_{l \rightarrow 3^+} \left[\frac{1}{z} \right]_{l-3}^1$$

$$= - \lim_{l \rightarrow 3^+} \left[1 - \frac{1}{l-3} \right] = -1 + \lim_{l \rightarrow 3^+} \frac{1}{l-3} = -1 + \frac{1}{3-3}$$

$= -1 + \infty = \infty$
diverges.

5



If f is continuous on the interval $[a, b]$, except for an infinite discontinuity at ' c ' in (a, b) , then the improper integral of ' f ' over the interval $[a, b]$ is defined as:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

The integral converges if both terms converge and diverges if either term diverges.

Example $\int_{-1}^8 x^{-1/3} dx$

$y = x^{-1/3} = \frac{1}{\sqrt[3]{x}}$
 $\therefore x \neq 0$

Hence an infinite discontinuity at $x=0$

$$= \int_{-1}^0 x^{-1/3} dx + \int_0^8 x^{-1/3} dx$$

$$= \lim_{l \rightarrow 0^-} \int_{-1}^l x^{-1/3} dx + \lim_{k \rightarrow 0^+} \int_k^8 x^{-1/3} dx$$

$$= \lim_{l \rightarrow 0^-} \left[\frac{x^{-1/3+1}}{-\frac{1}{3}+1} \right]_{-1}^l + \lim_{k \rightarrow 0^+} \left[\frac{x^{-1/3+1}}{-\frac{1}{3}+1} \right]_k^8$$

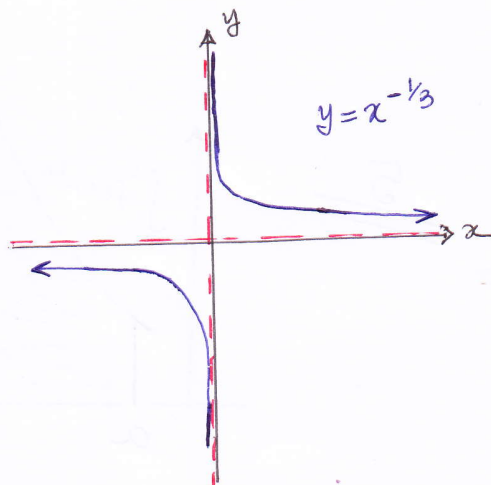
$$= \lim_{l \rightarrow 0^-} \left[\frac{x^{2/3}}{\frac{2}{3}} \right]_{-1}^l + \lim_{k \rightarrow 0^+} \left[\frac{x^{2/3}}{\frac{2}{3}} \right]_k^8$$

$$= \frac{3}{2} \lim_{l \rightarrow 0^-} \left[l^{2/3} - (-1)^{2/3} \right] + \frac{3}{2} \lim_{k \rightarrow 0^+} \left[8^{2/3} - k^{2/3} \right]$$

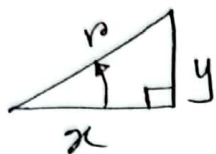
$$= \frac{3}{2} (0 - 1) + \frac{3}{2} (4 - 0)$$

$$= \frac{3}{2} (-1 + 4) = \frac{9}{2}$$

Converges



Ch 7.4. Integration by Trigonometric Substitution



$$x^2 + y^2 = r^2$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{y}{x} \quad \longleftrightarrow \quad \cot \theta = \frac{x}{y}$$

$$\sin \theta = \frac{y}{r} \quad \longleftrightarrow \quad \operatorname{cosec} \theta = \frac{r}{y}$$

$$\cos \theta = \frac{x}{r} \quad \longleftrightarrow \quad \sec \theta = \frac{r}{x}$$

$$\sin^2 \theta + \cos^2 \theta = 1 \rightarrow \begin{cases} 1 - \cos^2 \theta = \sin^2 \theta \\ 1 - \sin^2 \theta = \cos^2 \theta \end{cases}$$

$$\sec^2 \theta - \tan^2 \theta = 1 \rightarrow \begin{cases} 1 + \tan^2 \theta = \sec^2 \theta \end{cases}$$

$$\boxed{3} \int \frac{x^2}{\sqrt{16-x^2}} dx$$

$$\begin{cases} 1 - \sin^2 \theta = \cos^2 \theta \\ 1 - \cos^2 \theta = \sin^2 \theta \end{cases}$$

$$= \int \frac{x^2}{\sqrt{16(1 - \frac{x^2}{16})}} dx$$

$$= \int \frac{x^2}{4 \sqrt{1 - (\frac{x}{4})^2}} dx$$

$$= \frac{1}{4} \int \frac{16 \sin^2 \theta}{\sqrt{1 - \sin^2 \theta}} 4 \cos \theta d\theta$$

$$= 16 \int \frac{\sin^2 \theta \cos \theta}{\sqrt{\cos^2 \theta}} d\theta$$

$$= 16 \int \sin^2 \theta d\theta$$

$$= 16 \int \left(\frac{1}{2} (1 - \cos 2\theta) \right) d\theta$$

$$= 8 \int (1 - \cos 2\theta) d\theta$$

$$= 8 \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$= 8 \left[\sin^{-1} \frac{x}{4} - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$= 8 \left[\sin^{-1} \frac{x}{4} - \sin \theta \cos \theta \right] + C$$

$$= 8 \left[\sin^{-1} \frac{x}{4} - \frac{x}{4} \cdot \frac{\sqrt{16-x^2}}{4} \right] + C$$

$$= 8 \sin^{-1} \frac{x}{4} - \frac{x \sqrt{16-x^2}}{2} + C$$

$\boxed{2}$

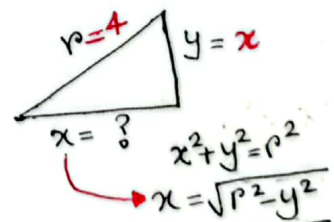
$$1 - \sin^2 \theta \leftrightarrow 1 - \frac{x^2}{16} = 1 - \left(\frac{x}{4} \right)^2$$

$$\text{let } \frac{x}{4} = \sin \theta$$

$$\frac{1}{4} dx = \cos \theta d\theta$$

$$dx = 4 \cos \theta d\theta$$

$$\begin{cases} \frac{x}{4} = \sin \theta \\ x = 4 \sin \theta \\ x^2 = 16 \sin^2 \theta \end{cases}$$



$$\frac{x}{4} = \sin \theta \quad \Rightarrow \quad x = \sqrt{r^2 - y^2} = \sqrt{16 - x^2}$$

$$\theta = \sin^{-1} \frac{x}{4}$$

$$y = r \sin \theta$$

$$\therefore \sin \theta = \frac{y}{r} = \frac{x}{4}$$

$$x = r \cos \theta$$

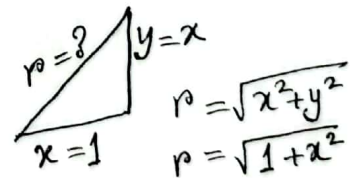
$$\therefore \cos \theta = \frac{x}{r} = \frac{\sqrt{16-x^2}}{4}$$

$$\begin{aligned}
\boxed{5} \quad & \int \frac{dx}{(4+4x^2)^2} \\
&= \int \frac{dx}{[4(1+x^2)]^2} \\
&= \frac{1}{16} \int \frac{dx}{[1+x^2]^2} \\
&= \frac{1}{16} \int \frac{\sec^2 \theta}{(1+\tan^2 \theta)^2} d\theta \\
&= \frac{1}{16} \int \frac{\sec^2 \theta}{\sec^4 \theta} d\theta \quad \rightarrow \frac{1}{\sec^2 \theta} \\
&= \frac{1}{16} \int \cos^2 \theta d\theta \quad \leftarrow \\
&= \frac{1}{16} \int \left(\frac{1+\cos 2\theta}{2} \right) d\theta \\
&= \frac{1}{32} \left[\theta + \frac{\sin 2\theta}{2} \right] \\
&= \frac{1}{32} \left[\theta + \frac{2\sin \theta \cos \theta}{2} \right] \\
&= \frac{1}{32} [\theta + \sin \theta \cos \theta] \\
&= \frac{1}{32} \left[\tan^{-1} x + \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} \right] + C \\
&= \frac{1}{32} \left[\tan^{-1} x + \frac{x}{1+x^2} \right] + C
\end{aligned}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\text{Let } \begin{cases} x = \tan \theta \\ dx = \sec^2 \theta d\theta \end{cases}$$

Our assumption
 $\tan \theta = x; \theta = \tan^{-1}(x)$
 but $\tan \theta = \frac{y}{x}$
 $\therefore \tan \theta = \frac{y}{x} = \frac{x}{1}$



$$\sin \theta = \frac{y}{r} = \frac{x}{\sqrt{1+x^2}}$$

$$\cos \theta = \frac{x}{r} = \frac{1}{\sqrt{1+x^2}}$$

$\boxed{3}$

$$\boxed{7} \int \frac{\sqrt{x^2-9}}{x} dx$$

$$= \int \frac{\sqrt{9(\frac{x^2}{9}-1)}}{x} dx$$

$$= 3 \int \frac{\sqrt{(\frac{x}{3})^2-1}}{x} dx$$

$$= 3 \int \frac{\sqrt{\sec^2\theta-1}}{3\sec\theta} (3\sec\theta\tan\theta d\theta)$$

$$= 3 \int \frac{\sqrt{\tan^2\theta}}{1} \tan\theta d\theta$$

$$= 3 \int \tan^2\theta$$

$$= 3 \int (\sec^2\theta - 1) d\theta$$

$$= 3 [\tan\theta - \theta] + C$$

$$= 3 \left[\frac{\sqrt{x^2-9}}{3} - \sec^{-1}\left(\frac{x}{3}\right) \right] + C$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$* \sec^2\theta - 1 = \tan^2\theta$$

$$\left\{ \begin{array}{l} u + \left\{ \frac{x^2}{9} - 1 = \sec^2\theta - 1 \right. \\ \left. \frac{x^2}{9} = \sec^2\theta \right. \\ \frac{x}{3} = \sec\theta \end{array} \right.$$

$$\frac{1}{3} dx = \sec\theta \tan\theta d\theta$$

$$dx = 3 \sec\theta \tan\theta d\theta$$

$$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\frac{x}{r}} = \frac{r}{x}$$

$$\sec\theta = \frac{r}{x} = \frac{x}{3} \text{ (Given)}$$

$$\begin{array}{l} r=x \\ x=3 \\ y=? \\ x^2+y^2=r^2 \\ y=\sqrt{r^2-x^2} \end{array}$$

$$\therefore y = \sqrt{x^2-9}$$

$$\tan\theta = \frac{y}{x} = \frac{\sqrt{x^2-9}}{3}$$

$$\sec\theta = \frac{x}{3} \text{ Given}$$

$$\theta = \sec^{-1}\left(\frac{x}{3}\right)$$

$\boxed{4}$

$$\boxed{23} \quad \int_{\sqrt{2}}^2 \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

$$= \int_{\pi/4}^{\pi/3} \frac{\sec \theta \tan \theta}{\sec^2 \theta \sqrt{\sec^2 \theta - 1}} d\theta$$

$$= \int_{\pi/4}^{\pi/3} \frac{\tan \theta}{\sec \theta \sqrt{\tan^2 \theta}} d\theta$$

$$= \int_{\pi/4}^{\pi/3} \frac{1}{\sec \theta} d\theta$$

$$= \int_{\pi/4}^{\pi/3} \cos \theta d\theta$$

$$= \left[\sin \theta \right]_{\pi/4}^{\pi/3}$$

$$= \sin \frac{\pi}{3} - \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}$$

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ \sec^2 \theta - 1 &= \tan^2 \theta \end{aligned}$$

$$\left\{ \sec^2 - 1 \leftrightarrow x^2 - 1 \right.$$

$$\text{let } \begin{cases} x = \sec \theta \\ x^2 = \sec^2 \theta \\ dx = \sec \theta \tan \theta d\theta \end{cases} \quad \boxed{\theta = \sec^{-1} x}$$

Limits

$$x = \sqrt{2} \rightarrow \theta = \sec^{-1} \sqrt{2} = \frac{\pi}{4}$$

$$x = 2 \rightarrow \theta = \sec^{-1} 2 = \frac{\pi}{3}$$

Limits

$$\sec^{-1} \sqrt{2} = \theta$$

$$\sec \theta = \sqrt{2}$$

$$\cos \theta = \frac{1}{\sec \theta} = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) = 45^\circ$$

$$\theta = \frac{\pi}{4} \text{ or } 45^\circ$$

$$\sec^{-1} 2 = \theta$$

$$\sec \theta = 2$$

$$\cos \theta = \frac{1}{\sec \theta} = \frac{1}{2}$$

$$\theta = \cos^{-1} \left(\frac{1}{2} \right) = 60^\circ$$

$$\theta = \frac{\pi}{3} \text{ or } 60^\circ$$

$\boxed{5}$

Examples

①

$$\begin{aligned} & \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4-x^2}} \\ &= \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4(1-\frac{x^2}{4})}} \\ &= \frac{1}{2} \int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{1-(\frac{x}{2})^2}} \\ &= \frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{2 \cos \theta d\theta}{4 \sin^2 \theta \sqrt{1-\sin^2 \theta}} \\ &= \frac{1}{4} \int_{\pi/6}^{\pi/4} \frac{\cos \theta}{\sin^2 \theta \sqrt{\cos^2 \theta}} d\theta \\ &= \frac{1}{4} \int_{\pi/6}^{\pi/4} \frac{1}{\sin^2 \theta} d\theta \\ &= \frac{1}{4} \int_{\pi/6}^{\pi/4} \operatorname{cosec}^2 \theta d\theta \\ &= -\frac{1}{4} [\cot \theta]_{\pi/6}^{\pi/4} \\ &= -\frac{1}{4} [\cot \frac{\pi}{4} - \cot \frac{\pi}{6}] \\ &= -\frac{1}{4} \left[\frac{1}{\tan \frac{\pi}{4}} - \frac{1}{\tan \frac{\pi}{6}} \right] \\ &= -\frac{1}{4} \left[\frac{1}{1} - \frac{1}{\frac{1}{\sqrt{3}}} \right] \\ &= -\frac{1}{4} [1 - \sqrt{3}] \\ &= \frac{\sqrt{3}-1}{4}. \end{aligned}$$

$$1 - \sin^2 \theta = \cos^2 \theta$$

let

$$\frac{x^2}{4} = \sin^2 \theta$$

$$\frac{x}{2} = \sin \theta$$

$$\frac{1}{2} dx = \cos \theta d\theta$$

$$dx = 2 \cos \theta d\theta$$

Limits
 $\frac{x}{2} = \sin \theta$

$$\theta = \sin^{-1} \frac{x}{2}$$

$$x=1 \rightarrow \theta = \sin^{-1} \left(\frac{1}{2} \right) = \pi/6$$

$$x=\sqrt{2} \rightarrow \theta = \sin^{-1} \left(\frac{\sqrt{2}}{2} \right)$$

$$= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \pi/4$$

6

$$\textcircled{\text{ii}} \int \frac{x}{x^2 - 4x + 8} dx$$

$$= \int \frac{x}{\underbrace{x^2 - 2 \cdot x \cdot 2 + 2^2}_{+4}} dx$$

$$= \int \frac{x}{(x-2)^2 + 4} dx$$

$$= \int \frac{(u+2)}{u^2 + 4} du$$

$$= \int \frac{u}{u^2 + 4} du + 2 \int \frac{1}{u^2 + 4} du$$

$$= \int \frac{\frac{1}{2} dz}{z} + 2 \int \frac{du}{4(\frac{u^2}{4} + 1)}$$

$$= \frac{1}{2} \ln z + \frac{1}{2} \int \frac{1}{1 + (\frac{u}{2})^2} du$$

$$= \frac{1}{2} \ln z + \frac{1}{2} \frac{\tan^{-1}(\frac{u}{2})}{\frac{1}{2}} + C$$

$$= \frac{1}{2} \ln(u^2 + 4) + \tan^{-1}\left(\frac{u}{2}\right) + C$$

$$= \frac{1}{2} \ln[(x-2)^2 + 4] + \tan^{-1}\left(\frac{x-2}{2}\right) + C$$

let
 $x-2 = u$
 $dx = du$
 $\therefore x = u+2$

let
 $u^2 + 4 = z$
 $2u du = dz$
 $u du = \frac{1}{2} dz$

Recall
 $\int \frac{1}{1+x^2} dx = \tan^{-1} x$

7

Integration Rational Function : Partial Fraction

Decomposition

$$i) \frac{3x-1}{(x-3)(x+4)} = \frac{A}{x-3} + \frac{B}{x+4} \rightarrow \frac{Ax^0}{x-3} + \frac{Bx^0}{x+4}$$

$$ii) \frac{5}{x(x^2-4)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-2}$$

$$iii) \frac{2x-3}{x^3-x^2} = \frac{2x-3}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$$

$$iv) \frac{x^2}{(x+2)^3} = \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{(x+2)^3} \quad \begin{array}{l} x^2 = x^2, x, 1 \\ x^5 = x^5, x^4, x^3, x^2, x, 1 \end{array}$$

$$v) \frac{1-x^2}{x^3(x^2+2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{Dx+E}{x^2+2}$$

$$vi) \frac{3x}{(x-1)(x^2+6)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+6}$$

$$vii) \frac{4x^3-x}{(x^2+5)^2} = \frac{Ax+B}{x^2+5} + \frac{Dx+E}{(x^2+5)^2}$$

$$viii) \frac{1-3x^1}{(x^3+3)(x-2)(x^2+1)^2} = \frac{Ax^2+Bx+C}{x^3+3} + \frac{D}{x-2} + \frac{Ex+F}{(x^2+1)} + \frac{Gx+H}{(x^2+1)^2}$$

Improper fraction : $\frac{3}{2}, \frac{2}{2}$ etc

$$\frac{3}{2} = 1\frac{1}{2}, \frac{2}{2} = 1$$

proper fractions : $\frac{1}{2}, -\frac{4}{7}$ etc

function of
Improper fraction :

$$\frac{x^3}{x^2+2}, \frac{x^2}{x^2+2} \text{ etc}$$

proper fraction :

$$\frac{x}{x^2+2} \text{ etc}$$

$$\int \frac{3x^1 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx$$

$$\begin{array}{r} 3x^2 + 1 \\ x^2 + x - 2 \overline{) 3x^1 + 3x^3 - 5x^2 + x - 1} \\ \underline{3x^1 + 3x^3 - 6x^2} \\ x^2 + x - 1 \\ \underline{x^2 + x - 2} \\ 1 \rightarrow R \end{array}$$

$$\begin{cases} x^2 + x - 2 \\ = x^2 + 2x - x - 2 \\ = x(x+2) - 1(x+2) \\ = (x+2)(x-1) \end{cases}$$

$$\rightarrow = \int \left[3x^2 + 1 + \left(\frac{1}{x^2 + x - 2} \right) \right] dx$$

$$= \frac{3x^3}{3} + x + \int \frac{1}{x^2 + x - 2} dx \quad \frac{1}{x^2 + x - 2} = \frac{1}{(x+2)(x-1)}$$

$$= \frac{3x^3}{3} + x + \int \left(\frac{A}{x+2} + \frac{B}{x-1} \right) dx$$

$$= \frac{3x^3}{3} + x + \int \left(\frac{-\frac{1}{3}}{x+2} + \frac{+\frac{1}{3}}{x-1} \right) dx$$

$$= \frac{3x^3}{3} + x - \frac{1}{3} \ln|x+2| + \frac{1}{3} \ln|x-1| + C$$

$$= x^3 + x + \frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + C$$

$$\frac{1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$$

$$1 = A(x-1) + B(x+2)$$

$$1 = Ax - A + Bx + 2B$$

Equating factors of like terms:

$$\begin{matrix} \text{"x"} & \text{"Constant"} \\ 0 = A + B & 1 = -A + 2B \end{matrix}$$

$$\therefore B = \frac{1}{3}, A = -\frac{1}{3}$$

$$(17) \int \frac{3x^2 - 10}{x^2 - 4x + 4} dx$$

$$= \int \left(3 + \frac{12x - 22}{x^2 - 4x + 4} \right) dx$$

$$= \int \left(3 + \frac{12x - 22}{(x-2)^2} \right) dx$$

$$= 3x + \int \frac{12x - 22}{(x-2)^2} dx$$

$$= 3x + \int \left(\frac{A}{(x-2)} + \frac{B}{(x-2)^2} \right) dx$$

$$= 3x + \int \left(\frac{12}{x-2} + \frac{2}{(x-2)^2} \right) dx$$

$$= 3x + 12 \ln|x-2| + \frac{2(x-2)^{-2+1}}{-2+1} + c$$

$$= 3x + 12 \ln|x-2| - \frac{2}{(x-2)} + c$$

$$\begin{array}{r} x^2 - 4x + 4 \overline{) 3x^2 + 0x - 10} \\ \underline{3x^2 - 12x + 12} \\ 12x - 22 \rightarrow R \end{array}$$

$$= 3 + \frac{12x - 22}{x^2 - 4x + 4}$$

$$\begin{aligned} x^2 - 4x + 4 &= x^2 - 2 \cdot x \cdot 2 + 2^2 \\ &= (x-2)^2 \end{aligned}$$

$$\frac{12x - 22}{(x-2)^2} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2}$$

$$12x - 22 = A(x-2) + B$$

$$12x - 22 = Ax - 2A + B$$

Equating factors of like terms

$$\begin{array}{l} \text{"x"} \\ \boxed{12 = A} \end{array} \quad \left\{ \begin{array}{l} \text{"Constant"} \\ -22 = -2A + B \\ \therefore \boxed{B = 2} \end{array} \right.$$

$$(29) \int \frac{2x^2-1}{(4x-1)(x^2+1)} dx$$

$$= \int \left(\frac{A}{4x-1} + \frac{Bx+C}{x^2+1} \right) dx$$

$$= \int \left(\frac{A}{4x-1} + \frac{Bx}{x^2+1} + \frac{C}{x^2+1} \right) dx$$

$$= \int \frac{-\frac{14}{17}}{4x-1} + \frac{\frac{12}{17}x}{x^2+1} + \frac{3/17}{x^2+1} dx$$

$$= -\frac{14}{17} \frac{\ln|4x-1|}{4} + \frac{12}{17} \int \frac{x}{x^2+1} dx + \frac{3}{17} \tan^{-1}x + C$$

$$= -\frac{14}{34} \ln|4x-1|$$

$$+ \frac{12}{17} \cdot \frac{1}{2} \ln(x^2+1) + \frac{3}{17} \tan^{-1}x + C$$

$$= -\frac{14}{34} \ln|4x-1| + \frac{6}{17} \ln(x^2+1) + \frac{3}{17} \tan^{-1}x + C$$

$$\frac{2x^2-1}{(4x-1)(x^2+1)} = \frac{A}{4x-1} + \frac{Bx+C}{x^2+1}$$

$$\frac{2x^2-1}{(4x-1)(x^2+1)} = \frac{A(x^2+1) + (Bx+C)(4x-1)}{(4x-1)(x^2+1)}$$

$$2x^2-1 = Ax^2+A+4Bx^2+4Cx-Bx-C$$

Equating factors of like terms:

$$\begin{array}{lcl} \text{"x"} & \text{"x"} & \text{"constant"} \\ 2 = A + 4B & 0 = 4C - B & -1 = A - C \end{array}$$

$$\textcircled{i} - \textcircled{iii} \rightarrow 3 = 4B + C \quad \textcircled{iv}$$

$$\textcircled{ii} \times 4 \rightarrow -4B + 16C = 0$$

$$\textcircled{iv} \times 1 \rightarrow 4B + C = 3$$

$$\begin{array}{r} \text{Add} \\ \hline 17C = 3 \\ \hline C = \frac{3}{17} \end{array}$$

$$\textcircled{iii} \quad A - C = -1$$

$$A - \frac{3}{17} = -1$$

$$\therefore A = -\frac{14}{17}$$

$$\textcircled{ii} \quad 4C - B = 0$$

$$4C = B$$

$$4\left(\frac{3}{17}\right) = B \quad \therefore B = \frac{12}{17}$$

$$\int \frac{x}{x^2+1} dx$$

$$= \int \frac{\frac{1}{2} du}{u}$$

$$= \frac{1}{2} \ln u$$

$$= \frac{1}{2} \ln(x^2+1)$$

$$\text{let } u = x^2+1$$

$$du = 2x dx$$

$$\frac{1}{2} du = x dx$$

Ch 7.3 Reduction Formula Anton's Calculus

$$\int \sin^n x dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int (\sin^{n-2} x) dx$$

$$\int \cos^n x dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int (\cos^{n-2} x) dx$$

$$\int \cos^5 x dx = \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \int \cos^3 x dx$$

$$= \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \int \cos^2 x \cos x dx$$

$$= \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \int (1 - \sin^2 x) \cos x dx$$

let $\sin x = u$
 $\cos x dx = du$

$$= \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \int (1 - u^2) du$$

$$= \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \left[u - \frac{u^3}{3} \right] + C$$

$$= \frac{1}{5} \cos^4 x \sin x + \frac{4}{5} \left[\sin x - \frac{\sin^3 x}{3} \right] + C$$

Alternatively

$$\int \cos^3 x dx = \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \int \cos x dx$$

$$= \frac{1}{3} \cos^2 x \sin x + \frac{2}{3} \sin x + C$$

using
reduction
formula

$$\textcircled{9} \int \sin^3 2x \cos^2 2x dx$$

$$= \int \sin^2 2x \cos^2 2x \sin 2x dx$$

$$= \int (1 - \cos^2 2x) \cos^2 2x \sin 2x dx$$

$$= \int (\cos^2 2x - \cos^4 2x) \sin 2x dx$$

$$= \int (u^2 - u^4) \left(-\frac{1}{2} du\right)$$

$$= -\frac{1}{2} \left[\frac{u^3}{3} - \frac{u^5}{5} \right] + C$$

$$= -\frac{1}{2} \left[\frac{\cos^3 2x}{3} - \frac{\cos^5 2x}{5} \right] + C$$

Let
 $\cos 2x = u$
 $-\sin 2x (2) dx = du$
 $\sin 2x = -\frac{1}{2} du$

$$\textcircled{10} \int \tan^5 x \sec^4 x dx$$

$$= \int \tan^5 x \sec^2 x \sec^2 x dx$$

$$= \int \tan^5 x (1 + \tan^2 x) \sec^2 x dx$$

$$= \int u^5 (1 + u^2) du$$

$$= \int (u^5 + u^7) du$$

$$= \frac{u^6}{6} + \frac{u^8}{8} + C$$

$$= \frac{\tan^6 x}{6} + \frac{\tan^8 x}{8} + C$$

Let $u = \tan x$
 $du = \sec^2 x dx$

Ch 7.2 Integration by Parts

$$\int uv = u \int v - \int (u' \int v)$$

Alternatively

$$\int u dv = uv - \int v du$$

$$\begin{aligned} u &= u(x) \\ v &= v(x) \end{aligned}$$

$$\textcircled{P} \int x \tan^2 x \, dx$$

$$\text{let } \begin{aligned} u &= x \\ v &= \tan^2 x \end{aligned}$$

$$= x \int \tan^2 x \, dx - \int \left\{ \frac{d}{dx} x \int \tan^2 x \, dx \right\} dx$$

$$= x \int (\sec^2 x - 1) \, dx - \int \{ 1 \int (\sec^2 x - 1) \, dx \} dx$$

$$= x [\tan x - x] - \int [\tan x - x] \, dx \quad \int \sec^2 x \, dx = \tan x$$

$$= x \tan x - x^2 - \left[\ln |\sec x| - \frac{x^2}{2} \right] + C \quad \begin{aligned} \int \tan x &= \ln |\sec x| \\ &= -\ln |\cos x| \end{aligned}$$

$$= x \tan x - \frac{x^2}{2} - \ln |\sec x| + C$$

$$\begin{aligned}
 \textcircled{ii} \quad & \int \cos(\ln x) dx \quad \leftarrow \begin{array}{l} \text{inner} \\ \text{outer} \\ \text{function} \end{array} \quad \begin{array}{l} \text{let } \ln x = z \\ \frac{1}{x} dx = dz \\ dx = x dz \\ = e^z dz \end{array} \quad \left. \begin{array}{l} \ln x = z \\ \log_e x = z \\ x = e^z \end{array} \right\} \\
 & = \int \cos z e^z dz \\
 & \text{let } u = e^z, \quad v = \cos z
 \end{aligned}$$

$$= e^z \int \cos z dz - \int \left(\frac{d}{dz} e^z \right) \int \cos z dz dz \quad \int uv = u \int v - \int (u' \int v)$$

$$= e^z \sin z - \int (e^z \sin z) dz$$

$$\text{let } u = e^z, \quad v = \sin z$$

$$= e^z \sin z - \left[e^z \int \sin z dz - \int \left(\frac{d}{dz} e^z \int \sin z \right) dz \right]$$

$$= e^z \sin z - \left[e^z (-\cos z) - \int e^z (-\cos z) dz \right]$$

$$= e^z \sin z + e^z \cos z - \int e^z \cos z dz$$

$$\therefore \int e^z \cos z dz = e^z \sin z + e^z \cos z - \int e^z \cos z dz$$

$$\Rightarrow 2 \int e^z \cos z dz = e^z \sin z + e^z \cos z$$

$$\Rightarrow \int e^z \cos z dz = \frac{1}{2} (e^z \sin z + e^z \cos z)$$

$$\Rightarrow \int \cos(\ln x) dx = \frac{1}{2} e^{\ln x} (\sin(\ln x) + \cos(\ln x))$$

$$= \frac{1}{2} x [\sin(\ln x) + \cos(\ln x)]$$

see Example (5) at page 494

Text Book Exercise Ch 7.2

$$\begin{aligned}
 & \int_1^3 \sqrt{x} \tan^{-1} \sqrt{x} \, dx \quad \int uv = u \int v - \int (u' \int v) \\
 & = \tan^{-1} \sqrt{x} \int x^{1/2} \, dx - \int \left[\frac{d}{dx} (\tan^{-1} \sqrt{x}) \int x^{1/2} \, dx \right] dx \quad \text{Let } u = \tan^{-1} \sqrt{x} \\
 & \quad \quad \quad v = \sqrt{x} = x^{1/2} \\
 & = \tan^{-1} \sqrt{x} \cdot \frac{x^{3/2}}{3/2} - \int \left[\left(\frac{1}{1+x} \right)^2 \cdot \frac{1}{2\sqrt{x}} \right] \cdot \frac{x^{3/2}}{3/2} \, dx \\
 & = \frac{2}{3} \tan^{-1} \sqrt{x} \cdot x^{3/2} - \int \left(\frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} \cdot \frac{x\sqrt{x}}{3/2} \right) dx \\
 & = \frac{2}{3} \tan^{-1} \sqrt{x} \cdot x^{3/2} - \int \left(\frac{1}{1+x} \cdot \frac{x}{2} \cdot \frac{2}{3} \right) dx \\
 & = \frac{2}{3} \tan^{-1} \sqrt{x} \cdot x^{3/2} - \int \frac{x}{3(1+x)} \, dx \\
 & = \frac{2}{3} x^{3/2} \tan^{-1} \sqrt{x} - \frac{1}{3} \int \frac{1+x-1}{1+x} \, dx \quad (\text{Algebraic manipulation}) \\
 & = \frac{2}{3} x^{3/2} \tan^{-1} \sqrt{x} - \frac{1}{3} \int \left(\frac{1+x}{1+x} - \frac{1}{1+x} \right) dx \\
 & = \frac{2}{3} x^{3/2} \tan^{-1} \sqrt{x} - \frac{1}{3} \int \left(1 - \frac{1}{1+x} \right) dx \\
 & = \left[\frac{2}{3} x^{3/2} \tan^{-1} \sqrt{x} - \frac{1}{3} (x - \ln(1+x)) \right]_1^3 \\
 & = \left[\frac{2}{3} x^{3/2} \tan^{-1} \sqrt{x} - \frac{x}{3} + \frac{\ln(1+x)}{3} \right]_1^3 \\
 & = \left[\frac{2}{3} \cdot 3^{3/2} \tan^{-1} \sqrt{3} - \frac{3}{3} + \frac{\ln(1+3)}{3} \right] - \left[\frac{2}{3} 1^{3/2} \tan^{-1} \sqrt{1} - \frac{1}{3} + \frac{\ln(1+1)}{3} \right] \\
 & = \frac{2}{3} \left(3^{3/2} \tan^{-1} \sqrt{3} - \tan^{-1} 1 \right) - \left(\frac{3}{3} - \frac{1}{3} \right) + \frac{1}{3} (\ln(4) - \ln(2)) \\
 & = \frac{2}{3} \left(3\sqrt{3} \frac{\pi}{3} - \frac{\pi}{4} \right) - \frac{2}{3} + \frac{1}{3} \ln \frac{4}{2} \\
 & = \frac{2\sqrt{3}}{3} \pi - \frac{\pi}{6} - \frac{2}{3} + \frac{1}{3} \ln 2 = \frac{1}{3} \left[2\sqrt{3} \pi - \frac{\pi}{2} - 2 + \ln 2 \right] \text{ Ans.}
 \end{aligned}$$



Beta Gamma Function

Beta function is also known as the Euler's integral defined by:

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx;$$
$$m > 0, n > 0.$$

It is a useful distribution to evaluate

- Laplace transformation
- Probability density function in statistics
- Quantum field theory in physics
- MATLAB programming

Gamma function (Γ) is an extension of the factorial function, with its argument shifted down by 1, to real and complex numbers.

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx = (n-1)!$$

Gamma Beta function can be related to each other: $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$

$$\text{Trigonometric Function: } \int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}.$$

$$\text{OR: } \int_0^{\pi/2} 2\sin^{2x-1}(t)\cos^{2y-1}(t)dt = \beta(x, y)$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$2. (ix) \int_0^\infty \frac{1}{1+x^4} dx$$

Trigonometric Substitution:

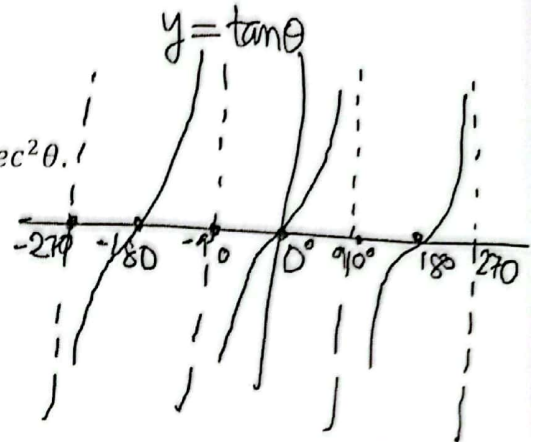
$$x^2 = \tan \theta; \therefore x^4 = \tan^2 \theta; 1 + x^4 = 1 + \tan^2 \theta = \sec^2 \theta.$$

$$x = \sqrt{\tan \theta}$$

$$dx = \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta$$

$$\text{Limits: } x = 0 \Rightarrow \theta = 0$$

$$x = \infty \Rightarrow \theta = \frac{\pi}{2}$$



$$\star \tan 0^\circ = \frac{\sin 0^\circ}{\cos 0^\circ} = \frac{0}{1} = 0$$

$$\star \tan 90^\circ = \frac{\sin 90^\circ}{\cos 90^\circ} = \frac{1}{0} = \infty$$

$$\int_0^\infty \frac{1}{1+x^4} dx$$

$$= \int_0^{\pi/2} \frac{1}{1+\tan^2 \theta} \cdot \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta$$

$$= \int_0^{\pi/2} \frac{\sec^2 \theta}{\sec^2 \theta} \cdot \frac{1}{2\sqrt{\tan \theta}} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{1}{2} \int_0^{\pi/2} \sqrt{\frac{\cos \theta}{\sin \theta}} d\theta = \frac{1}{2} \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{1/2} \theta d\theta$$

Alternative Method

$$= \frac{1}{4} \int_0^{\pi/2} 2 \sin^{-1/2} \theta \cos^{1/2} \theta d\theta$$

$$= \frac{1}{4} \beta\left(\frac{1}{4}, \frac{3}{4}\right)$$

$$\begin{matrix} \uparrow & \uparrow \\ x & y \end{matrix}$$

$$2y-1 = \frac{1}{2}$$

$$2y = \frac{1}{2} + 1 = \frac{3}{2}$$

$$y = \frac{3}{4}$$

$$2x-1 = -\frac{1}{2}$$

$$2x = -\frac{1}{2} + 1$$

$$2x = \frac{1}{2}$$

$$x = \frac{1}{4}$$

$$\int_0^{\pi/2} 2 \sin^{2x-1}(\theta) \cos^{2y-1}(\theta) d\theta = \beta(x, y)$$

$$\text{OR} \quad = \frac{1}{2} \frac{\Gamma\left(\frac{-\frac{1}{2}+1}{2}\right) \Gamma\left(\frac{\frac{1}{2}+1}{2}\right)}{2\Gamma\left(\frac{-\frac{1}{2}+\frac{1}{2}+2}{2}\right)}$$

$$= \frac{1}{4} \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)}{\Gamma(1)}$$

$$= \frac{1}{4} \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right)}{\Gamma\left(\frac{1}{4} + \frac{3}{4}\right)}$$

$$= \frac{1}{4} \beta\left(\frac{1}{4}, \frac{3}{4}\right).$$

$$\frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = \beta(m, n)$$

$$\int_0^{\pi/2} \sin^p x \cos^q x \, dx = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$$

$$3(i) \int_0^{\pi} \sin^5 \theta \cos^4 \theta \, d\theta$$

$$= 2 \int_0^{\pi/2} \sin^5 \theta \cos^4 \theta \, d\theta$$

$$= 2 \frac{\Gamma\left(\frac{5+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{5+4+2}{2}\right)}$$

$$= 2 \frac{\Gamma(3) \Gamma\left(\frac{5}{2}\right)}{2\Gamma\left(\frac{11}{2}\right)}$$

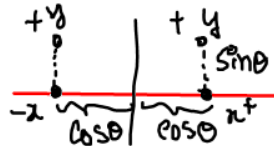
$$= \frac{\Gamma(3) \Gamma\left(\frac{5}{2}\right)}{\Gamma\left(3+\frac{5}{2}\right)} = \beta\left(3, \frac{5}{2}\right).$$

$$\int_0^{2a} f(x) \, dx = 2 \int_0^a f(x) \, dx \text{ when } f(2a-x) = -f(x)$$

$$\therefore \int_0^{\pi} \sin^5 \theta \cos^4 \theta \, d\theta = 2 \int_0^{\pi/2} \sin^5 \theta \cos^4 \theta \, d\theta \text{ if } f(\pi-\theta) = -f(\theta).$$

$$f(\theta) = \sin^5 \theta \cos^4 \theta$$

$$l.s. = f(\pi-\theta) = \sin^5(\pi-\theta) \cos^4(\pi-\theta) = (\sin^5 \theta)(-\cos^4 \theta)$$



$$= -\sin^5 \theta \cos^4 \theta = -f(\theta)$$

$$\therefore \int_0^{\pi} \sin^5 \theta \cos^4 \theta \, d\theta = 2 \int_0^{\pi/2} \sin^5 \theta \cos^4 \theta \, d\theta. \quad 3(ii) \text{ is similar to } 3(i)$$

$$\text{Alternative Method: } \int_0^{\pi/2} 2\sin^{2x-1}(t)\cos^{2y-1}(t) \, dt = \beta(x, y)$$

$$3(i) \int_0^{\pi} \sin^5 \theta \cos^4 \theta \, d\theta$$

$$= 2 \int_0^{\pi/2} \sin^5 \theta \cos^4 \theta \, d\theta$$

$$= \int_0^{\pi/2} 2\sin^5 \theta \cos^4 \theta \, d\theta$$

$$= \int_0^{\pi/2} 2\sin^{2x-1}(\theta)\cos^{2y-1}(\theta) \, d\theta, \text{ where } 2x-1=5, 2y-1=4 \text{ hence } x=3, y=\frac{5}{2}$$

$$= \beta(x, y) = \beta\left(3, \frac{5}{2}\right)$$

Beta Function

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$m > 0, n > 0$

Exercises

$$2^{iii}) \int_0^1 \frac{1}{\sqrt{1-x^3}} dx$$

$$= \int_0^1 (1-x^3)^{-1/2} dx$$

$$= \int_0^1 \frac{1}{3} z^{-2/3} (1-z)^{-1/2} dz$$

$\xrightarrow{m-1} \quad \xrightarrow{n-1}$

$$= \frac{1}{3} \int_0^1 z^{1/3-1} (1-z)^{1/2-1} dz$$

$$= \frac{1}{3} \beta\left(\frac{1}{3}, \frac{1}{2}\right)$$

$$x^3 = z$$

$$3x^2 dx = dz$$

$$dx = \frac{1}{3x^2} dz$$

$$= \frac{1}{3} z^{-2/3} dz$$

limits

$$x=0 \rightarrow z=0$$

$$x=1 \rightarrow z=1$$

$$\rightarrow m-1 = -2/3$$

$$m = 1 - 2/3 = 1/3$$

$$\rightarrow n-1 = -1/2$$

$$n = \frac{1}{2}$$

$$2vi) \int_0^a y^7 \sqrt{a^4 - y^4} dy$$

$\sqrt{a^4(1-y^4)} = a^2 \sqrt{1-y^4}$
 $a^4 y^4 \rightarrow$ factor out a^4

Let $y^4 = a^4 z^4$
 $y = az$
 $dy = a dz$

$$= \int_0^1 a^7 z^7 \sqrt{a^4 - a^4 z^4} a dz$$

limits

$$y=0 \Rightarrow az=0$$

$$z=0$$

$$y=a \Rightarrow az=a$$

$$z=1$$

$$= a^8 \int_0^1 z^7 a^2 \sqrt{1-z^4} dz$$

$$= a^{10} \int_0^1 z^7 (1-z^4)^{1/2} dz$$

$$= a^{10} \int_0^1 z^4 \cdot z^3 (1-z^4)^{1/2} dz$$

let $z^4 = x$
 $4z^3 dz = dx$
 $z^3 dz = \frac{1}{4} dx$

$$= a^{10} \int_0^1 x \frac{1}{4} (1-x)^{1/2} dx$$

limits

$$z=0 \rightarrow x=0$$

$$z=1 \rightarrow x=1$$

$$= \frac{a^{10}}{4} \int_0^1 x^1 (1-x)^{1/2} dx$$

$$= \frac{a^{10}}{4} \int_0^1 x^{2-1} (1-x)^{3/2-1} dx$$

$$n-1 = \frac{1}{2}$$

$$n = \frac{1}{2} + 1 = \frac{3}{2}$$

$$= \frac{a^{10}}{4} \beta(2, 3/2)$$

2

Gamma Function

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx ; n > 0$$

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} ; \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$m > 0, n > 0$

$$1e) \int_0^4 x^{3/2} (4-x)^{5/2} dx$$

$$x = 4y$$

$$dx = 4dy$$

$$= \int_0^1 4^{3/2} y^{3/2} (4-4y)^{5/2} 4dy$$

limits

$$x=0 \rightarrow y=0$$

$$x=4 \rightarrow y=1$$

$$= 4^{3/2+1} \int_0^1 y^{3/2} 4^{5/2} (1-y)^{5/2} dy$$

$$= 4^{\frac{3}{2}+1+\frac{5}{2}} \int_0^1 y^{\overset{m-1}{3/2}} (1-y)^{\overset{n-1}{5/2}} dy$$

$$m-1 = 3/2$$

$$m = 3/2 + 1 = 5/2$$

$$= 4^{\frac{10}{2}} \int_0^1 y^{\overset{5/2-1}{5/2}} (1-y)^{\overset{7/2-1}{7/2}} dy$$

$$n-1 = 5/2$$

$$n = 5/2 + 1 = 7/2$$

$$= 4^5 \beta\left(\frac{5}{2}, \frac{7}{2}\right)$$

$$= 1024 \frac{\Gamma(\frac{5}{2}) \Gamma(\frac{7}{2})}{\Gamma(\frac{5}{2} + \frac{7}{2})} = 1024 \frac{\Gamma(\frac{5}{2}) \Gamma(\frac{7}{2})}{\Gamma(6)}$$

3

$$1(i) \int_0^b y^5 \sqrt{b^2 - y^2} dy$$

$$= \int_0^1 b^5 z^5 \sqrt{b^2 - b^2 z^2} b dz$$

$$= b^6 \int_0^1 z^5 b \sqrt{1 - z^2} dz$$

$$= b^7 \int_0^1 z^5 (1 - z^2)^{1/2} dz$$

$$= b^7 \int_0^1 z \cdot z^4 (1 - z^2)^{1/2} dz$$

$$= b^7 \int_0^1 \frac{1}{2} \cdot x^2 (1 - x)^{1/2} dx$$

$$= \frac{b^7}{2} \int_0^1 x^{3-1} (1 - x)^{3/2-1} dx$$

$$= \frac{b^7}{2} \beta(3, 3/2)$$

$$= \frac{b^7}{2} \frac{\Gamma(3) \cdot \Gamma(3/2)}{\Gamma(3 + 3/2)}$$

$$= \frac{b^7}{2} \frac{\Gamma(3) \cdot \Gamma(3/2)}{\Gamma(9/2)}$$

$$\text{let } y = bz$$

$$dy = b dz$$

limits

$$y=0 \rightarrow z=0$$

$$y=b \rightarrow z=1$$

$$z^2 = x \rightarrow z^4 = x^2$$

$$2z dz = dx$$

$$z dz = \frac{1}{2} dx$$

limits:

$$z=0 \rightarrow x=0$$

$$z=1 \rightarrow x=1$$

$$1 \text{ vii) } \int_0^{\infty} x^6 e^{-3x} dx$$

$$= \int_0^{\infty} \left(\frac{z}{3}\right)^6 e^{-z} \frac{1}{3} dz$$

$$= \frac{1}{3^7} \int_0^{\infty} e^{-z} z^{7-1} dz$$

$$= \frac{1}{3^7} \Gamma(7)$$

$$1 \text{ ix) } \int_0^{\infty} \sqrt{x} e^{-x^2} dx$$

$$= \int_0^{\infty} z^{1/4} e^{-z} \frac{1}{2\sqrt{z}} dz$$

$$= \frac{1}{2} \int_0^{\infty} z^{1/4 - 1/2} e^{-z} dz$$

$$= \frac{1}{2} \int_0^{\infty} z^{(-1/4)} e^{-z} dz$$

$$= \frac{1}{2} \int_0^{\infty} e^{-z} z^{\frac{3}{4}-1} dz$$

$$= \frac{1}{2} \Gamma\left(\frac{3}{4}\right)$$

$$3x = z \rightarrow \frac{z}{3} = x$$

$$3dx = dz$$

$$dx = \frac{1}{3} dz$$

limits:

$$x=0 \rightarrow z=0$$

$$x=\infty \rightarrow z=\infty$$

$$x^2 = z \rightarrow x = \sqrt{z}$$

$$2x dx = dz$$

$$dx = \frac{1}{2x} dz$$

$$= \frac{1}{2\sqrt{z}} dz$$

limits

$$x=0 \rightarrow z=0$$

$$x=\infty \rightarrow z=\infty$$

$$n-1 = -\frac{1}{4}$$

$$n = -\frac{1}{4} + 1 = \frac{3}{4}$$

$$\sqrt{x \ln \frac{1}{x}} = x^{1/2} \sqrt{\ln \left(\frac{1}{x}\right)}$$

$$1 \text{ x)} \int_0^1 \frac{1}{\sqrt{x \ln \left(\frac{1}{x}\right)}} dx$$

$$= \int_{-\infty}^0 \frac{-e^{-z} dz}{\sqrt{e^{-z} z}}$$

$$= - \int_{-\infty}^0 \frac{e^{-z} dz}{e^{-z/2} z^{1/2}}$$

$$= \int_0^{\infty} e^{-z - (-z/2)} z^{-1/2} dz$$

$$= \int_0^{\infty} e^{-z/2} z^{-1/2} dz \xrightarrow{\text{let } \frac{z}{2} = y}$$

$$= \int_0^{\infty} e^{-y} (2y)^{-1/2} 2 dy$$

$$= 2^{-1/2+1} \int_0^{\infty} e^{-y} y^{-1/2} dy$$

$$= \sqrt{2} \int_0^{\infty} e^{-y} y^{\frac{1}{2}-1} dy$$

$$= \sqrt{2} \Gamma\left(\frac{1}{2}\right)$$

$$1(x11) \int_0^1 \left(1 - \frac{1}{x}\right)^{1/3} dx$$

$$= \int_0^1 \left(\frac{x-1}{x}\right)^{1/3} dx = \int_0^1 - \left(\frac{1-x}{x}\right)^{1/3} dx$$

$$= - \int_0^1 x^{-1/3} (1-x)^{1/3} dx$$

∴ continue

[6]

$$\text{let } \ln\left(\frac{1}{x}\right) = z$$

$$\ln 1 - \ln x = z$$

$$0 - \ln x = z$$

$$\ln x = -z$$

$$\log_e x = -z$$

$$x = e^{-z}$$

$$dx = -e^{-z} dz$$

$$\text{limits: } x=0 \rightarrow z=\infty \rightarrow \ln\left(\frac{1}{0}\right) = \ln \infty = \infty$$

$$x=1 \rightarrow z=0 \rightarrow \ln\left(\frac{1}{1}\right) = \ln 1 = 0$$

$$\frac{z}{2} = y$$

$$z = 2y$$

$$dz = 2dy$$

limits

$$z=0 \rightarrow y = \frac{0}{2} = 0$$

$$z=\infty \rightarrow y = \frac{\infty}{2} = \infty$$

Formulae: $\boxed{1} \int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2 \Gamma(\frac{p+q+2}{2})}$
or $\boxed{2} \int_0^{\pi/2} 2 \sin^{2x-1}(t) \cos^{2y-1}(t) dt = \beta(x, y)$

3 iii) $\int_0^{\pi/6} \sin^2 6x \cos^4 3x dx$

$= \int_0^{\pi/6} (\sin 2 \cdot 3x)^2 \cos^4(3x) dx$

$\sin^2 6x = (\sin 6x)^2$
 $= (\sin 2 \cdot 3x)^2$
 $= (2 \sin 3x \cos 3x)^2$

$\sin^2 \theta = (\sin \theta)^2$

$\sin^2 6x = (\sin 6x)^2$
 $= (\sin 2 \cdot 3x)^2$

$= \int_0^{\pi/6} (2 \sin 3x \cdot \cos 3x)^2 \cos^4 3x dx$

$= 4 \int_0^{\pi/6} \sin^2 3x \cos^2 3x \cos^4 3x dx$

$= 4 \int_0^{\pi/6} \sin^2 3x \cos^6 3x dx$

let $3x = z$
 $3 dx = dz$
 $dx = \frac{1}{3} dz$

$= 4 \cdot \frac{1}{3} \int_0^{\pi/2} \sin^2 z \cos^6 z dz$

$= \frac{4}{3} \frac{\Gamma(\frac{2+1}{2}) \Gamma(\frac{6+1}{2})}{2 \Gamma(\frac{2+6+2}{2})}$

$= \frac{4}{3} \frac{\Gamma(3/2) \Gamma(7/2)}{2 \Gamma(5)} = \frac{2}{3} \frac{\Gamma(3/2) \Gamma(7/2)}{\Gamma(5)}$

limits:
 $x=0 \rightarrow z=0$
 $x=\pi/6 \rightarrow z=\pi/2$

Alternative Method:

$= \frac{2}{3} \int_0^{\pi/2} 2 \sin^2 z \cos^6 z dz$
 $= \frac{2}{3} \beta(x, y) = \frac{2}{3} \beta(\frac{3}{2}, \frac{7}{2})$

$2x-1=2$
 $2x=2+1$
 $x=3/2$
 $2y-1=6$
 $2y=7$
 $y=7/2$

$\frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = \beta(m, n)$
 $\beta(\frac{3}{2}, \frac{7}{2}) = \frac{\Gamma(3/2) \Gamma(7/2)}{\Gamma(3/2+7/2)}$

MAT 120

Beta and Gamma Functions

1. Evaluate in terms of gamma function: (WEEK 2)

$$\begin{aligned} & \text{(i)} \int_0^4 x^{3/2} (4-x)^{5/2} dx, \quad \text{(ii)} \int_0^b y^5 \sqrt{b^2 - y^2} dy, \quad \text{(iii)} \int_0^\infty e^{-t^2} dt, \\ & \text{(iv)} \int_0^\infty x^5 e^{-4x} dx, \quad \text{(v)} \int_0^\infty e^{-y^2} y^5 dy, \quad \text{(vi)} \int_0^\infty e^{-x^2} dx, \quad \text{(vii)} \int_0^\infty x^6 e^{-3x} dx, \\ & \text{(viii)} \int_0^\infty e^{-x^2} x^9 dx, \quad \text{(ix)} \int_0^\infty \sqrt{x} e^{-x^2} dx, \quad \text{(x)} \int_0^1 \frac{x^3}{\sqrt{1-x^3}} dx, \\ & \text{(xi)} \int_0^1 \frac{1}{\sqrt{x \ln(1/x)}} dx, \quad \text{(xii)} \int_0^1 \left(1 - \frac{1}{x}\right)^{1/3} dx. \end{aligned}$$

2. Evaluate in terms of beta function: (WEEK 3)

$$\begin{aligned} & \text{(i)} \int_0^1 \frac{x^2}{\sqrt{1-x}} dx, \quad \text{(ii)} \int_0^1 x^7 (1-x)^3 dx, \quad \text{(iii)} \int_0^1 \frac{1}{\sqrt{1-x^3}} dx, \\ & \text{(iv)} \int_0^1 (1-x)^{1/2} x^3 dx, \quad \text{(v)} \int_0^1 x^{5/2} (1-x)^{3/2} dx, \\ & \text{(vi)} \int_0^a y^7 \sqrt{a^4 - y^4} dy, \quad \text{(vii)} \int_0^4 y^3 \sqrt{64 - y^3} dy, \\ & \text{(viii)} \int_0^1 x^2 (1-x^3)^{3/2} dx, \quad \text{(ix)} \int_0^\infty \frac{1}{1+x^4} dx. \end{aligned}$$

3. Evaluate the following integrals: (WEEK 3)

$$\begin{aligned} & \text{(i)} \int_0^\pi \sin^5 \theta \cos^4 \theta d\theta, \quad \text{(ii)} \int_0^\pi \sin^6 \theta \cos^7 \theta d\theta, \\ & \text{(iii)} \int_0^{\pi/6} \sin^2 6x \cos^4 3x dx, \quad \text{(iv)} \int_0^{\pi/4} \sin^2 4\theta \cos^3 2\theta d\theta, \\ & \text{(v)} \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta, \quad \text{(vi)} \int_0^{\pi/8} \sin^2 8x \cos^4 4x dx. \end{aligned}$$

Formula

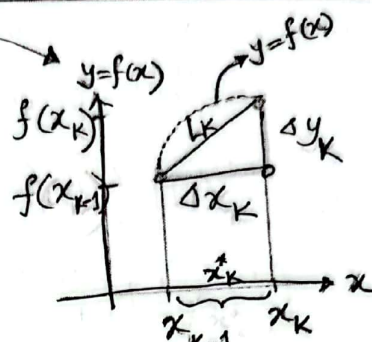
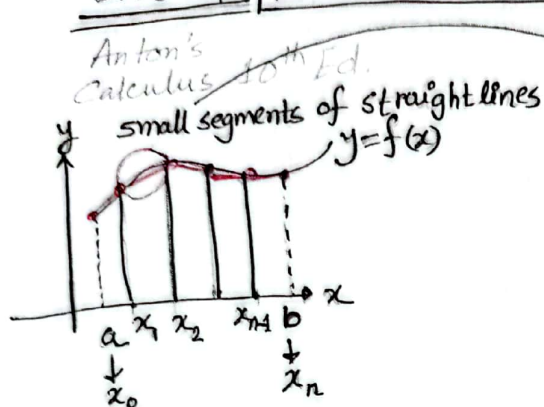
1. $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad \text{where } n > 0$
2. $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad \text{where } m > 0, n > 0.$
3. $\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2 \Gamma(\frac{p+q+2}{2})}$
4. $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
5. $\Gamma(n) = (n-1)!$
6. $\Gamma(n+1) = n\Gamma(n) = n!$
7. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$
8. $\Gamma(1) = 1$
9. $\Gamma(\frac{p}{2}) = (\frac{p}{2}-1)(\frac{p}{2}-2)(\frac{p}{2}-3)\dots\dots\dots\frac{1}{2} \cdot \Gamma(\frac{1}{2})$
10. $\Gamma(n) = (n-1)(n-2)\dots\dots\dots 3.2.1$
11. $\int_0^{\pi/2} 2 \sin^{2x-1}(t) \cos^{2y-1}(t) dt = \beta(x, y)$

Gamma Beta Distribution Solution

1. (i) $8.4.32 \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{\Gamma(6)}$
 (ii) $\frac{b^7}{2} \frac{\Gamma(3)\Gamma(\frac{3}{2})}{\Gamma(\frac{9}{2})}$
 (iii) $\frac{1}{2} (1/2)$
 (iv) $\frac{1}{4^6} (6)$
 (v) $\frac{1}{2} (3)$
 (vii) $\frac{1}{3^7} (7)$
 (viii) $\frac{1}{2} (5)$
 (ix) $\frac{1}{2} (3/4)$
 (x) $\frac{1}{3} \frac{\Gamma(\frac{4}{3})\Gamma(\frac{1}{3})}{\Gamma(\frac{4}{3}+\frac{1}{3})}$
 (xi) $\sqrt{2} (1/2)$
 (xii) $-\frac{\Gamma(\frac{2}{3})\Gamma(\frac{4}{3})}{\Gamma(2)}$
2. (i) $\beta(3, \frac{1}{2})$
 (ii) $\beta(8, 4)$
 (iii) $\frac{1}{3}\beta(1/3, \frac{1}{2})$
 (iv) $\beta(4, 3/2)$
 (v) $\beta(7/2, 5/2)$
 (vi) $\frac{\alpha^{10}}{4}\beta(2, 3/2)$
 (vii) $4^4 \cdot 8 \cdot \frac{1}{3}\beta(4/3, 3/2)$
 (viii) $\frac{1}{3}\beta(1, 5/2)$
 (ix) $\frac{1}{2} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})}{2\Gamma(1)} = \frac{1}{4} \beta(\frac{1}{4}, \frac{3}{4})$
3. (i) $\frac{\Gamma(3)\Gamma(\frac{5}{2})}{\Gamma(\frac{11}{2})}$ (vi) $\frac{\Gamma(\frac{3}{2})\Gamma(\frac{7}{2})}{2\Gamma(5)}$
 (ii) $\frac{\Gamma(\frac{7}{2})\Gamma(4)}{\Gamma(\frac{15}{2})}$
 (iii) $\frac{4}{3} \frac{\Gamma(\frac{3}{2})\Gamma(\frac{7}{2})}{2\Gamma(5)}$
 (iv) $\frac{\Gamma(\frac{3}{2})\Gamma(3)}{\Gamma(\frac{9}{2})}$
 (v) $\frac{\Gamma(\frac{5}{2})\Gamma(\frac{3}{2})}{2\Gamma(4)}$

MAT 120 Week 4

Ch6.4 LENGTH of PLANE CURVE



$$x_k^* \in [x_{k-1}, x_k]$$

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \text{ by Pythagorean Theorem}$$

Single segment of the curve

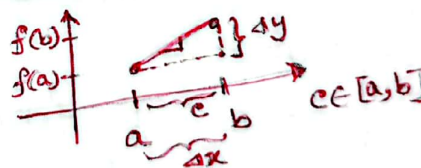
$$L_k = \sqrt{(\Delta x_k)^2 + [f(x_k) - f(x_{k-1})]^2}$$

n many segment of the curve

Recall Mean Value Theorem

There exist c between a & b such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

The arc length is the measure of the distance along the curved line making up the arc. It is longer than the straight line distance between its end points.



There exist a number x_k^* between x_{k-1} and x_k such that $\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} = f'(x_k^*)$

$$\Rightarrow f(x_k) - f(x_{k-1}) = f'(x_k^*) \Delta x_k \quad (2)$$

Substitute (2) into (1)

$$L = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + [f'(x_k^*)]^2 (\Delta x_k)^2} = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 (1 + [f'(x_k^*)]^2)}$$

Riemann sum to Riemann Integral

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

1

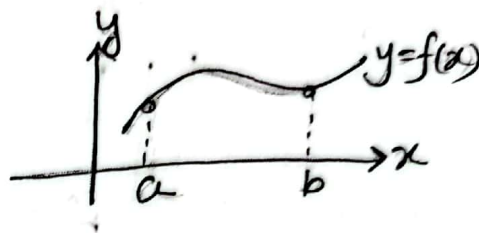
A smooth function is a function that has continuous derivatives up to some desired order over some domain.

Definition

If $y=f(x)$ is a smooth curve on the interval $[a,b]$, the arc length L of this curve $[a,b]$ is defined as

$$L = \int_a^b \sqrt{1+[f'(x)]^2} dx$$

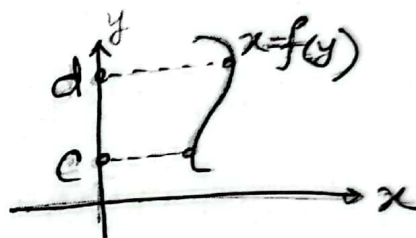
$$= \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2} dx$$



$x=f(y)$
 f is continuous on $[c,d]$.

$$L = \int_c^d \sqrt{1+[f'(y)]^2} dy$$

$$= \int_c^d \sqrt{1+\left(\frac{dx}{dy}\right)^2} dy$$



Exercise: (Text Book)

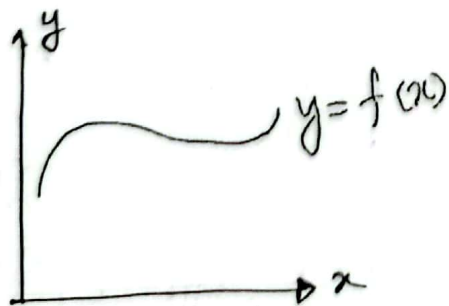
Find the exact arc length of the curve over the stated interval:

③ $y = 3x^{3/2} - 1$ from $x=0$ to $x=1$

$$y=f(x) = 3x^{3/2} - 1$$

$$f'(x) = 3 \cdot \frac{3}{2} x^{3/2-1} - 0 = \frac{9}{2} x^{1/2}$$

$$L = \int_0^1 \sqrt{1+\left(\frac{9}{2} x^{1/2}\right)^2} dx = \int_0^1 \sqrt{1+\frac{81}{4} x} dx$$



$$= \frac{4}{81} \int_1^{85/4} \sqrt{z} dz$$

$$= \frac{4}{81} \left[\frac{z^{3/2}}{3/2} \right]_1^{85/4}$$

$$= \frac{4}{81} \cdot \frac{2}{3} \left[\left(\frac{85}{4}\right)^{3/2} - (1)^{3/2} \right]$$

$$= 3.19$$

Let

$$1+\frac{81}{4}x = z$$

$$\frac{81}{4}dx = dz$$

$$dx = \frac{4}{81}dz$$

Limits

$$x=0 \rightarrow z=1$$

$$x=1 \rightarrow z=\frac{85}{4}$$

5) $f(x) = y = x^{2/3}$; from $x=1$ to $x=8$

$$f'(x) = \frac{2}{3} x^{-1/3}$$

$$[f'(x)]^2 = \frac{4}{9} x^{-2/3}$$

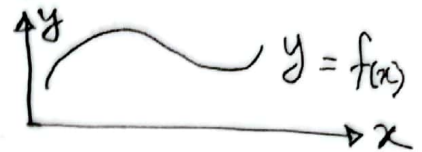
$$L = \int_1^8 \sqrt{1 + \frac{4}{9} x^{-2/3}} dx$$

$$= \int_1^8 \sqrt{\frac{9x^{2/3} + 4}{9x^{2/3}}} dx$$

$$= \frac{1}{3} \int_1^8 \frac{\sqrt{9x^{2/3} + 4}}{x^{1/3}} dx$$

$$= \frac{1}{3} \int_{13}^{40} \sqrt{z} \left(\frac{1}{6} dz \right)$$

⋮
continue.



$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$\frac{4}{9x^{2/3}}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Let

$$9x^{2/3} + 4 = z$$

$$9 \cdot \frac{2}{3} x^{-1/3} dx = dz$$

$$\frac{6}{x^{1/3}} dx = dz$$

$$\left(\frac{1}{x^{1/3}} dx \right) = \left(\frac{1}{6} dz \right)$$

limits:

$$x=1 \rightarrow z=13$$

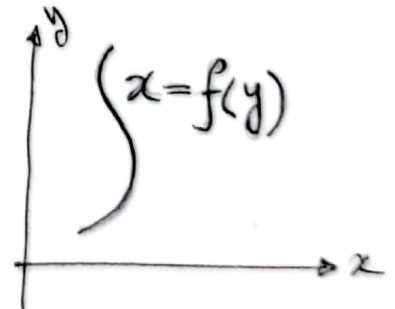
$$x=8 \rightarrow z=40$$

7) $24xy = y^4 + 48$, $\left\{ \begin{array}{l} y=c \text{ to } y=d \\ y=2 \text{ to } y=4 \end{array} \right\}$

$$x = \frac{y^4 + 48}{24y} \rightarrow f(y)$$

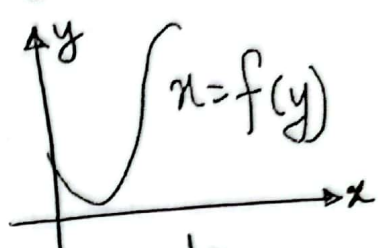
$$\frac{dx}{dy} = \frac{24y(4y^3) - (y^4 + 48)24}{(24y)^2}$$

$$= \frac{24[4y^4 - y^4 - 48]}{24^2 y^2} = \frac{3y^4 - 48}{24y^2} = \frac{y^4 - 16}{8y^2}$$



3

$$\begin{aligned}
 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} &= \sqrt{1 + \left(\frac{y^4 - 16}{8y^2}\right)^2} \\
 &= \sqrt{1 + \frac{(y^4 - 16)^2}{64y^4}} \\
 &= \sqrt{\frac{64y^4 + y^8 - 32y^4 + 256}{64y^4}} \\
 &= \sqrt{\frac{32y^4 + y^8 + 256}{64y^4}} \\
 &= \frac{\sqrt{(y^4)^2 + 2y^4 \cdot 16 + 16^2}}{8y^2} \\
 &= \frac{\sqrt{(y^4 + 16)^2}}{8y^2} = \frac{y^4 + 16}{8y^2}
 \end{aligned}$$

$$L = \int_c^d \sqrt{1 + [f'(y)]^2} dy$$


$f'(y) = \frac{dx}{dy}$

$$L = \int_2^4 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= \int_2^4 \frac{y^4 + 16}{8y^2} dy$$

$$= \int_2^4 \left(\frac{y^2}{8} + \frac{2}{y^2} \right) dy$$

$$= \left[\frac{y^3}{24} + \frac{2y^{-2+1}}{-2+1} \right]_2^4 = \left[\frac{y^3}{24} - 2y^{-1} \right]_2^4$$

$$= \frac{64-8}{24} - 2[(4)^{-1} - (2)^{-1}]$$

$$\begin{aligned}
 &= \frac{56}{24} - 2\left[-\frac{1}{4} - \frac{1}{2}\right] = \frac{7}{3} - 2\left[-\frac{1}{4}\right] \\
 &= \frac{7}{3} + \frac{1}{2} = \frac{14+3}{6} = \frac{17}{6}
 \end{aligned}$$

4

Parametric Eqn

Ex

$$x = x(t), \quad y = y(t)$$

Consider

- ① $x = 2t, \quad y = t^2$ } • pair of these together called parametric eqns
• $t = \text{parameter}$

- ② $x = \sin\theta + 2, \quad y = \cos\theta - 3$ } → parametric eqn
 $\theta = \text{parameter.}$

Say, if $x = 2$

$$\text{then } 2 = \sin\theta + 2$$

$$0 = \sin\theta$$

$$\theta = \sin^{-1}(0) = 0, \pi, 2\pi,$$

if $\theta = 0$ then $y = \cos(0) - 3 = -2$
and so on.

Evaluate Cartesian Form:

eqn with x, y and parameter eliminated

case ① $x = 2t, \quad y = t^2$ } ①

from ① $t = \frac{x}{2}$

Substitute ① into ② we get:

$$y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4} \rightarrow \text{This is the Cartesian form}$$

case ② $x = \sin\theta + 2$ } ①

$$\sin\theta = x - 2$$
 } ①

$$\sin^2\theta = (x - 2)^2$$
 } ①

$$y = \cos\theta - 3$$
 } ②

$$\cos\theta = y + 3$$
 } ②

$$\cos^2\theta = (y + 3)^2$$
 } ②

$$(\text{①} + \text{②}) \Rightarrow \sin^2\theta + \cos^2\theta = (x - 2)^2 + (y + 3)^2$$

$$1 = (x - 2)^2 + (y + 3)^2 \rightarrow \theta \text{ is eliminated}$$

$$1^2 = (x - h)^2 + (y - k)^2 \rightarrow r = 1$$

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = -2t \quad \left\{ \text{Center } (2, -3) \right.$$

Differentiation:

$$x = x(t), \quad y = y(t)$$

Consider $x = 2t, \quad y = t^2$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= 2t \cdot \frac{1}{2} = t$$

110
differentiation

Coordinate
Geometry

110

For Parametric curves:

If the number of segments of the curve represented by the parametric eqn say $x = x(t)$, $y = y(t)$, ($a \leq t \leq b$) is traced more than once as t increases from a to b , and if $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are continuous functions for $t \in [a, b]$, then the arc length L of the curve is given by $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Ex $x(\theta) = \cos \theta$, $y(\theta) = \sin \theta$, $0 \leq \theta \leq 2\pi$

$$\frac{dx}{d\theta} = -\sin \theta$$

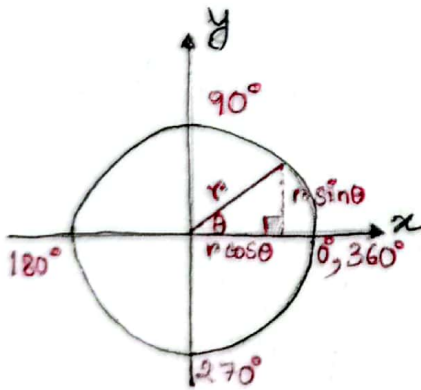
$$\frac{dy}{d\theta} = \cos \theta$$

$$L = \int_0^{2\pi} \sqrt{\sin^2 \theta + \cos^2 \theta} d\theta$$

$$= \int_0^{2\pi} d\theta$$

$$= [\theta]_0^{2\pi} = 2\pi$$

↓
arc length of full circle



Find the arc length of the parametric curve

$$(27) \quad x = \frac{1}{3}t^3, \quad y = \frac{1}{2}t^2, \quad 0 \leq t \leq 1$$

$$\frac{dx}{dt} = \frac{1}{3} \cdot 3t^2 = t^2, \quad \frac{dy}{dt} = \frac{1}{2} \cdot 2t = t$$

$$L = \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= \int_0^1 \sqrt{t^4 + t^2} dt$$

$$= \int_0^1 \sqrt{t^2(t^2+1)} dt$$

$$= \int_0^1 t \sqrt{t^2+1} dt$$

$$= \int_1^2 \sqrt{z} \cdot \frac{1}{2} dz$$

$$= \frac{1}{2} \left[\frac{z^{3/2}}{3/2} \right]_1^2 = \frac{1}{2} \cdot \frac{2}{3} \left[(2)^{3/2} - (1)^{3/2} \right] = \frac{1}{3} (2\sqrt{2} - 1)$$

$$\begin{aligned} \text{let } t^2+1 &= z \\ 2t dt &= dz \\ t dt &= \frac{1}{2} dz \\ t=0 &\rightarrow z=1 \\ t=1 &\rightarrow z=2 \end{aligned}$$

$$(29) \quad x = \cos 2t, \quad y = \sin 2t, \quad 0 \leq t \leq \pi/2$$

$$\frac{dx}{dt} = -2\sin 2t, \quad \frac{dy}{dt} = 2\cos 2t$$

$$L = \int_0^{\pi/2} \sqrt{(-2\sin 2t)^2 + (2\cos 2t)^2} dt$$

$$= \int_0^{\pi/2} \sqrt{4\sin^2 2t + 4\cos^2 2t} dt$$

$$= \int_0^{\pi/2} \sqrt{4(\sin^2 2t + \cos^2 2t)} dt$$

$$= 2 \int_0^{\pi/2} \sqrt{1} dt = 2 [t]_0^{\pi/2} = \pi$$

Ch 6.5

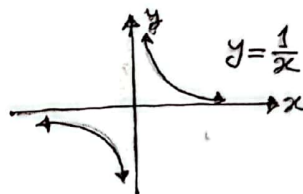
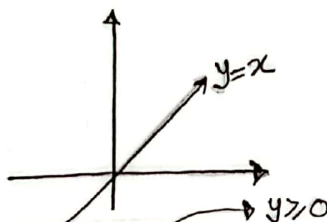
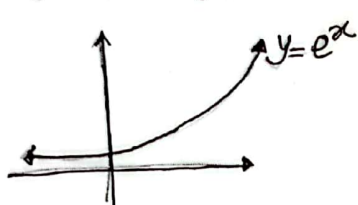
Anton's
Calculus
10th Ed.

AREA of a SURFACE of REVOLUTION

A smooth function has continuous derivatives up to some desired order over some domain.

A function said to be smooth over a restricted interval I such as (a, b) or $[a, b]$.

$y = e^x$, $y = x$ etc are examples of smooth functions.

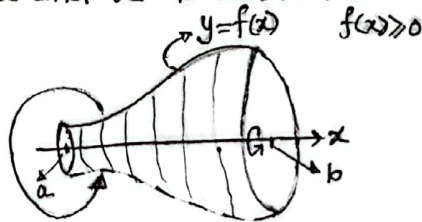


If f is a smooth non-negative function on $[a, b]$, then the surface area S of the surface of revolution that is generated by revolving the portion

$y = \frac{1}{x}$ is a piecewise smooth function. It is discontinuous at $x = 0$.

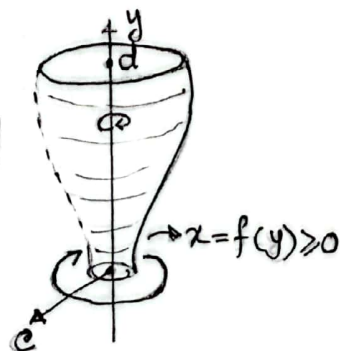
of the curve $y = f(x)$ between $x = a$ and $x = b$ about the x -axis is defined as:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$



About the y -axis is defined as:

$$S = \int_c^d 2\pi f(y) \sqrt{1 + [f'(y)]^2} dy$$



Find the area of the surface generated by revolving given curve about the x-axis:

$$y = \sqrt{x} - \frac{1}{3}x^{3/2}; \quad 1 \leq x \leq 3$$

$$f(x) = \sqrt{x} - \frac{1}{3}x^{3/2}$$

$$S = 2\pi f(x) \sqrt{1 + [f'(x)]^2}$$

$$f'(x) = \frac{dy}{dx} = \frac{1}{2\sqrt{x}} - \frac{1}{3} \cdot \frac{3}{2} x^{1/2}$$

$$= \frac{1}{2\sqrt{x}} - \frac{1}{2}x^{1/2}$$

$$= \frac{1}{2} \left(\frac{1}{\sqrt{x}} - \sqrt{x} \right) = \frac{1}{2} \left(\frac{1-x}{\sqrt{x}} \right)$$

$$S = \int_1^3 2\pi \left(\sqrt{x} - \frac{1}{3}x^{3/2} \right) \sqrt{1 + \left[\frac{1}{2} \left(\frac{1-x}{\sqrt{x}} \right) \right]^2} dx$$

$$= 2\pi \int_1^3 \left(\sqrt{x} - \frac{1}{3}x^{3/2} \right) \sqrt{1 + \frac{(1-x)^2}{4x}} dx$$

$$= 2\pi \int_1^3 \frac{3\sqrt{x} - x\sqrt{x}}{3} \sqrt{\frac{4x+1-2x+x^2}{4x}} dx$$

$$= 2\pi \int_1^3 \frac{\sqrt{x}(3-x)}{3} \sqrt{\frac{x^2+2x+1}{4x}} dx$$

$$= \frac{2\pi}{3} \int_1^3 \sqrt{x}(3-x) \frac{\sqrt{(x+1)^2}}{2\sqrt{x}} dx$$

$$= \frac{\pi}{3} \int_1^3 (3-x)(x+1) dx$$

$$= \frac{\pi}{3} \int_1^3 (3x - x^2 + 3 - x) dx$$

$$= \frac{\pi}{3} \int_1^3 (2x - x^2 + 3) dx$$

$$\begin{aligned}
&= \frac{\pi}{3} \left[\frac{2x^2}{2} - \frac{x^3}{3} + 3x \right]_1^3 \\
&= \frac{\pi}{3} \left[x^2 - \frac{x^3}{3} + 3x \right]_1^3 \\
&= \frac{\pi}{3} \left[3^2 - \frac{3^3}{3} + 3(3) - 1^2 + \frac{1^3}{3} - 3(1) \right] \\
&= \frac{\pi}{3} \left[9 - 9 + 9 - 1 + \frac{1}{3} - 3 \right] \\
&= \frac{\pi}{3} \left[5 + \frac{1}{3} \right] = \frac{\pi}{3} \left[\frac{16}{3} \right] = \frac{16\pi}{9}
\end{aligned}$$

Find the area of the surface generated by revolving the given curve about the y -axis.

$$x = 9y + 1 \quad ; \quad 0 \leq y \leq 2$$

$$\frac{dx}{dy} = 9$$

$$S = \int_0^2 2\pi f(y) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= 2\pi \int_0^2 (9y+1) \sqrt{1 + (9)^2} dy$$

$$= 2\pi \int_0^2 (9y+1) \sqrt{82} dy$$

$$= 2\sqrt{82} \pi \int_0^2 (9y+1) dy = 2\sqrt{82} \pi \left[\frac{9y^2}{2} + y \right]_0^2$$

$$= 2\sqrt{82} \pi \left[\frac{36}{2} + 2 - 0 \right]$$

$$= 40\sqrt{82} \pi$$

$$x = \sqrt{9-y^2} ; -2 \leq y \leq 2$$

$$f'(y) = \frac{dx}{dy} = \frac{1}{2\sqrt{9-y^2}} (-2y) = \frac{-y}{\sqrt{9-y^2}}$$

$$S = \int_{-2}^2 2\pi \sqrt{9-y^2} \sqrt{1 + \left(\frac{-y}{\sqrt{9-y^2}}\right)^2} dy$$

$$= 2\pi \int_{-2}^2 \sqrt{9-y^2} \sqrt{1 + \frac{y^2}{9-y^2}} dy$$

$$= 2\pi \int_{-2}^2 \sqrt{9-y^2} \sqrt{\frac{9-y^2+y^2}{9-y^2}} dy$$

$$= 2\pi \int_{-2}^2 \cancel{\sqrt{9-y^2}} \frac{\overset{\rightarrow 3}{\sqrt{9}}}{\cancel{\sqrt{9-y^2}}} dy$$

$$= 6\pi \int_{-2}^2 dy$$

$$= 6\pi [y]_{-2}^2 = 24\pi$$

[4]